How Transformers Learn to Plan via Multi-Token Prediction

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Abstract

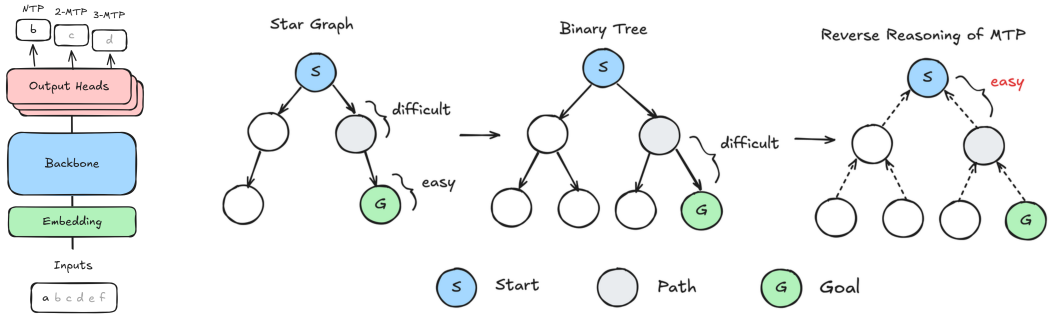
While next-token prediction (NTP) has been the standard objective for training language models, it often struggles to capture global structure in reasoning tasks. Multi-token prediction (MTP) has recently emerged as a promising alternative, yet its underlying mechanisms remain poorly understood. In this paper, we study how MTP facilitates reasoning, with a focus on planning. Empirically, we show that MTP consistently outperforms NTP on both synthetic graph path-finding tasks and more realistic reasoning benchmarks, such as Countdown and boolean satisfiability problems. Theoretically, we analyze a simplified two-layer Transformer on a star graph task. We prove that MTP induces a two-stage *reverse reasoning* process: the model first attends to the end node and then reconstructs the path by tracing intermediate nodes backward. This behavior arises from a gradient decoupling property of MTP, which provides a cleaner training signal compared to NTP. Ultimately, our results highlight how multi-token objectives inherently bias optimization toward robust and interpretable reasoning circuits.

1 Introduction

Recently, large language models (LLMs) have demonstrated a range of emerging abilities as they scale, with reasoning becoming one of the most critical capabilities (Ke et al., 2025). Reasoning enables language models to go beyond simple knowledge retrieval, allowing them to actively process and infer solutions to new problems. Despite these advances, how such reasoning abilities emerge in language models remain poorly understood. Recent theoretical advances focused on the role of chain-of-thought (Feng et al., 2023; Kim & Suzuki, 2025; Huang et al., 2025) and architecture design (Gatmiry et al., 2024; Saunshi et al., 2025) in enabling reasoning. However, a more fundamental factor may lie in the modeling paradigm itself, particularly the training objective that underlies language model training.

Next-token prediction (NTP) with teacher forcing has long been the standard objective for training language models, where models are optimized to produce the next token given all preceding ground-truth context tokens. Yet, it is limited in capturing long-term dependencies (Qi et al., 2020) and prone to overfitting to local patterns (Bachmann & Nagarajan, 2024). To address these limitations, multi-token prediction (MTP) (Gloeckle et al., 2024) has recently been proposed. Instead of predicting only the next token, it predicts multiple future tokens in parallel. MTP substantially improves performance on math and code benchmarks that require complex reasoning, and has been widely adopted in leading models such as DeepSeek-V3 (Liu et al., 2024). Understanding how MTP leads to such improvements is crucial for uncovering the mechanisms underlying the reasoning capabilities of modern language models.

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(a) (b)

Figure 1: **Overview of the MTP architecture and its advantage in reasoning tasks.** (a) The MTP architecture. Building upon Gloeckle et al. (2024), our MTP employs a shared backbone with multiple independent output heads to predict several future tokens simultaneously. (b) Illustration of Star Graph and Binary Tree. In standard forward planning, early steps can be difficult due to a large search space. MTP facilitates "reverse reasoning" by looking ahead to the goal (G), transforming complex forward searches into easy backward steps.

In this work, we study how MTP facilitates reasoning in language models, with a focus on *planning*[1], which requires considering future steps before generating the current output. Among MTP variants, we adopt the formulation of Gloeckle et al. (2024), which predicts multiple future tokens using parallel heads. Note that our primary focus is on the reasoning performance rather than inference efficiency. Though the model is trained to predict multiple tokens, it generates only a single token at each step using the first head during inference.

We begin with empirical observations on synthetic graph path-finding tasks, which are widely regarded as a standard abstraction for studying reasoning (Kim et al., 2025; Ye et al., 2025). Given the graph structure and start/end nodes, the model is required to generate the full path. We find that MTP consistently outperforms NTP on these tasks. We first study the star graph (see Figure 1b), a directed graph in which multiple paths originate from the start node, with one leading to the target. In a star graph, only the first step requires a non-trivial decision. Thus, under NTP with teacher forcing, the model tends to follow edges based on previously revealed nodes in the prefix, a phenomenon known as the Clever Hans cheat (Bachmann & Nagarajan, 2024). To remove this shortcut, we introduce a more complex task, the *binary tree*, in which decisions must be made at every step. Notably, MTP continues to outperform NTP, indicating that its advantage cannot be explained solely by disabling the Clever Hans cheat. We evaluate on more realistic tasks, including Countdown (Gandhi et al., 2024) and boolean satisfiability problems, where MTP still succeeds.

We next delve deeper into the underlying mechanisms behind the success of MTP on planning tasks, with a focus on the star graph. For simplicity, we study a two-layer disentangled Transformer (Friedman et al., 2023) on the 2-path 3-node star graph. Our theory reveals a two-stage *reverse reasoning* process in models trained with MTP: the model first attends to the end node, then retrieves intermediate nodes by following edges that point to it. Intuitively, solving the star graph is easier when starting from the end node, which extends to the binary tree task. Importantly, MTP modifies only the training objective, while the inference process remains the same as NTP, indicating that reverse reasoning arises from optimization differences. Under NTP, learning signals from different layers are entangled, making it difficult for the model to discover the patterns needed for planning. In contrast, MTP provides an isolated training signal, allowing the first layer to attend to the end node and the second layer to recover the intermediate node through simple edge matching.

[1]Reasoning refers to the ability to perform logical or compositional inference, whereas planning is a specific form of reasoning that requires anticipating future steps and computing a global solution before generating outputs.

Technical novelty. We are the first to formally analyze the convergence dynamics of MTP and establish its difference from NTP, extending beyond prior results (Zhong et al., 2025). The key is the *gradient decoupling* property of the MTP head, which provides an isolated training signal to the first layer. This property enables the emergence of *reverse reasoning* mechanism. Our technique might have independent interest to learning theory.

2 Related Work

Understanding of MTP. Despite the empirical success of MTP, a rigorous theoretical understanding of its underlying mechanisms remains unclear. Recent work (Zhong et al., 2025) studied how MTP enhances planning on synthetic tasks. However, its analysis is largely qualitative and does not provide a formal comparison with NTP. Moreover, its theoretical setup considers a simplified shared-head variant of MTP, rather than the standard independent-head architecture introduced by Gloeckle et al. (2024). *In contrast*, our work provides a theoretical comparison between NTP and MTP under the more general independent-head architecture. We mathematically unpack how the gradient decoupling property of MTP naturally induces a two-stage *reverse reasoning* process on the star graph, providing a theoretical explanation for its empirical advantage over NTP. For an extended discussion of related work on MTP, we refer to Appendix A.

Theories of Transformers. Extensive theoretical studies have been conducted to characterize the mechanisms of Transformer architectures, covering aspects such as in-context learning (Zhang et al., 2024; Chen et al., 2024b; Wei et al., 2026), chain of thought (Kim & Suzuki, 2025; Huang et al., 2025; Wen et al., 2025), induction head (Nichani et al., 2024; Chen et al., 2024a), benign overfitting (Jiang et al., 2024; Magen et al., 2024), attention sink (Guo et al., 2025) and compositional reasoning (Wang et al., 2025b; 2026). Within this landscape, disentangled transformers (Friedman et al., 2023; Nichani et al., 2024; Chen et al., 2024a; Wei et al., 2026) have gained significant attention, as they decouple the intertwined features in the residual stream to facilitate a more tractable analytical framework. Following this paradigm, we adopt a disentangled Transformer as the structural foundation for our study.

3 Preliminaries

Notation and Basic Setups. We use $[N] = \{1, \ldots, N\}$ to denote the vertex set and $\{e_i\}_{i=1}^N$ for the standard basis of $\mathbb{R}^N$. For a sequence of length T, we write $x_{1:T} = (x_1, \ldots, x_T)$ and use $[\cdot, :]$ to denote a row slice of a matrix. The softmax function $\sigma : \mathbb{R}^k \to \mathbb{R}^k$ is defined by $\sigma(v)_i = \exp(v_i) / \sum_j \exp(v_j)$, applied row-wise. For a softmax distribution s, we write $J(s) := \mathrm{diag}(s) - ss^\top$ for its Jacobian. We use I_T for the $T \times T$ identity matrix and $L \in \mathbb{R}^{T \times T}$ for the strictly lower shift matrix with $L_{i,i-1} = 1$ and zero elsewhere.

Training Objectives. Let $x_{1:T} = (x_1, \ldots, x_T)$ be an input sequence. The standard NTP objective minimizes the negative log-likelihood of each token given its full causal prefix:

$$\mathcal{L}_{\mathrm{NTP}}(\theta) = -\frac{1}{T} \sum_{t=1}^{T} \log p_\theta \left(x_t \mid x_{1:t-1} \right). \tag{1}$$

MTP with lookahead k instead predicts the next k tokens in parallel from the same prefix, averaging the cross-entropy over all k heads:

$$\mathcal{L}_{\mathrm{MTP}}^{(k)}(\theta) = -\frac{1}{T} \sum_{t=1}^{T} \frac{1}{k} \sum_{m=1}^{k} \log p_{\theta,m} \left(x_{t+m} \mid x_{1:t} \right), \tag{2}$$

where $p_{\theta,m}$ denotes the m-th prediction head. In both objectives, training uses teacher forcing: the ground-truth prefix $x_{1:t}$ is supplied as context regardless of the model's own predictions. Note that NTP is a special case of MTP with $k = 1$.

Architectural Design. We adopt a parallel architecture (Gloeckle et al., 2024) as illustrated in Figure 1a, which utilizes a shared transformer backbone capped with multiple independent

linear output heads for simultaneous MTP. This contrasts with the sequential structure employed by Liu et al. (2024). Our choice is motivated by the goal of deconstructing the underlying mechanisms of MTP rather than maximizing absolute performance. By prioritizing a streamlined architecture, we ensure better theoretical tractability while still effectively capturing the empirical advantages of MTP over standard NTP.

Main Experimental Setups. Across all experiments, models are trained using the MTP objective with the architecture described above. However, to evaluate the core reasoning and generative capabilities acquired during training, all models are evaluated using standard NTP inference. For statistical reliability, all results are averaged over 3 independent runs for each data point. Details of the experimental configurations are provided in Appendix E.1.

4 Planning Emerges with Multi-Token Prediction

In this section, we provide empirical evidence that MTP enables the planning abilities in Transformers. On complex planning tasks, including star graph (Bachmann & Nagarajan, 2024), countdown (Gandhi et al., 2024), and boolean satisfiability, MTP consistently outperforms NTP across a wide range of model and data scales.

Star Graph: NTP Fails, but MTP Succeeds. We begin with one of the simplest planning tasks, the star graph. A star graph is a directed graph with v_{start} as the central node and at least two paths originating from it. One of these paths leads to the target node v_{end}. Given the graph and the start/end nodes, the goal is to find the path from v_{start} to v_{end}. See Figure 1b for an example. In this task, the model cannot fully rely on previously generated tokens to determine the next token; instead, it must combine information across multiple path segments to infer reachability. Bachmann & Nagarajan (2024) showed that NTP fails on the star graph task. We replicate this finding and further evaluate MTP on the same task.

As shown in Figure 2a-b, we train Transformers with MTP on star graphs with two paths, each of five nodes. Specifically, we compare MTPs with different lookahead steps k; when $k = 1$, MTP naturally reduces to NTP. Following Kaplan et al. (2020); Hoffmann et al. (2022), we evaluate MTP and NTP under both data-scaling and parameter-scaling setups. In Figure 2a, as the training data size increases, the path-finding accuracy of models trained with MTP ($k \geq 2$) surpasses that of NTP. Surprisingly, even $k = 2$, which predicts only one additional future token beyond NTP, achieves 100% accuracy when trained on 0.5M samples, whereas NTP remains at 50% accuracy across different training data sizes. Similar trends are observed under parameter scaling in Figure 2b. These findings indicate the potential of MTP for planning tasks over NTP.

Binary Tree: No Clever Hans Cheat, yet MTP Still Wins. Bachmann & Nagarajan (2024) attributed the failure of NTP on star graph tasks to the *Clever Hans Cheat* phenomenon. This phenomenon arises from teacher-forced training, where the model is given the prefix of the ground-truth answer while predicting the next token. The model may exploit trivial correlations between previously revealed answer tokens and the next token, instead of solving the underlying task. In the star graph task, when predicting node v_i, the model already sees previous node v_{i-1} as input. Since every intermediate node has only one outgoing edge, the model can simply follow the edge from v_{i-1} to predict v_i. Consequently, the model fits the training data but never learns the true path-planning mechanism.

Bachmann & Nagarajan (2024) further introduced *teacherless* training to eliminate the Clever Hans Cheat by replacing the ground-truth prefix with dummy tokens. The model must predict multiple future tokens without relying on previously revealed answers. Conceptually, this teacherless objective is closely related to modern MTP. Then, a natural question arises:

Does MTP enable better planning simply by bypassing the cheating mechanism?

To answer this question, we introduce a new task, the *binary tree* problem, which extends the two-path star graph design. See Figure 1b as an illustration. In the star graph, only the start node branches into two paths, whereas the binary tree branches at every intermediate node. Thus, each prediction along the correct path cannot rely on previously revealed nodes, preventing the same cheating as in the star graph.

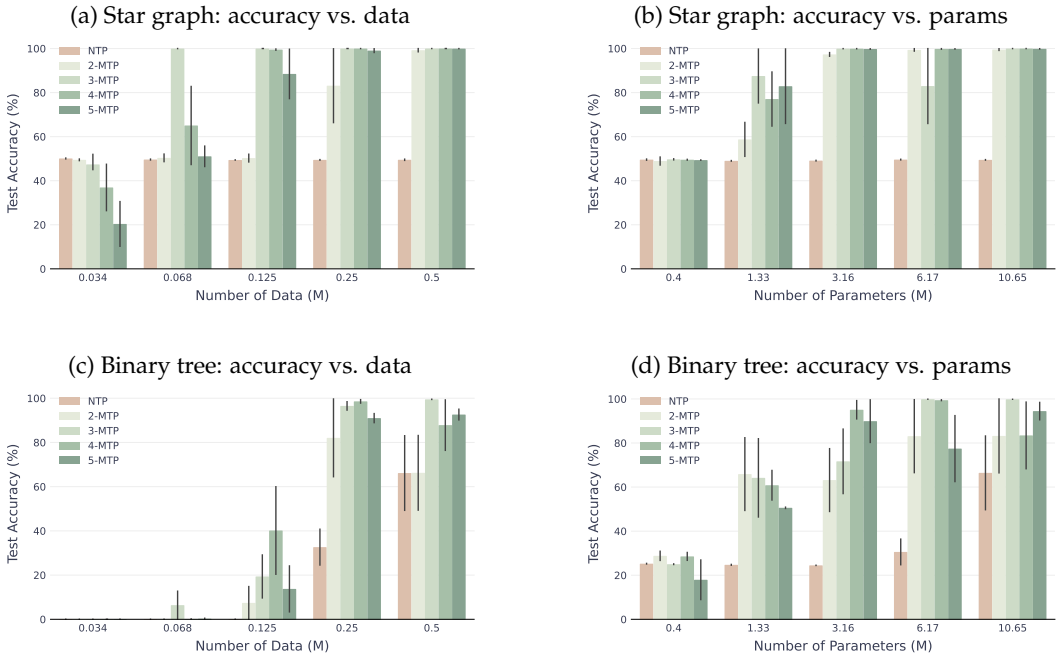


Figure 2: Path-finding accuracy of MTP versus NTP under varying data and parameter scales. **(a, b)** 2-path 5-node star graph: test accuracy as a function of training data size (16M model) and parameter size (0.5M data). **(c, d)** 5-node binary tree: test accuracy as a function of data size (16M model) and parameter size (1M data). Error bars represent 90% confidence intervals calculated from three independent replicates ($n = 3$). Notably, while NTP stagnates at 50% accuracy on star graph, it begins to learn and scale on binary trees. Across both topologies, MTP consistently demonstrates superior planning abilities.

Now we evaluate MTP and NTP on the binary tree tasks. As before, we adopt both data-scaling and parameter-scaling setups. In Figure 2c-d, models trained with MTP still outperform those trained with NTP in most settings, especially when data or model capacity is limited. Unlike the star graph experiments, the performance of NTP improves as the data size or model size increases. It is clear that, without the Clever Hans cheat, NTP begins to learn planning, while MTP still demonstrates stronger planning ability. We conclude that MTP facilitates planning not merely by bypassing the cheating mechanism; deeper mechanisms are involved, which we discuss further in Section 5.

More Planning Tasks: Countdown and SAT. To further validate planning abilities of MTP, we consider two more realistic planning tasks: Countdown and Boolean satisfiability (SAT). Both tasks require computing the full solution internally before generating the first token, thereby demanding extensive planning over a large number of combinations. See Appendix B for illustrations of these tasks.

Countdown. Countdown is a generalized version of the Game of 24. The goal is to use the given numbers and arithmetic operations to reach a target number. Yao et al. (2023) showed even frontier models such as GPT-4 struggle with Countdown problems.

SAT. SAT is a classic problem in computer science and was proven to be NP-complete (Cook, 1971). The goal is to find a Boolean assignment to variables that satisfies all clauses of a given propositional formula.

In Table 1, we present the performance of models trained with MTP and NTP on these two tasks. Evidently, models trained with MTP achieve consistently higher test accuracy than those trained with NTP.

TASK	NTP	2-MTP	3-MTP	4-MTP	5-MTP	6-MTP	7-MTP
Countdown	60.27	60.36	63.20	62.09	62.75	63.17	**64.93**
3-SAT	10.40	28.17	69.50	63.10	82.83	82.00	**87.47**

Table 1: Test accuracy of models trained with NTP versus MTP configurations on Countdown and 3-SAT tasks. Models trained with MTP achieve consistently higher test accuracy than the NTP baseline across both tasks.

In summary, we highlight two key findings. First, MTP outperforms NTP on star graph tasks, not merely by disabling the Clever Hans Cheat; deeper mechanisms are involved. Second, MTP consistently enables better planning in Transformers across both graph path-finding and more realistic tasks.

5 Mechanisms of Planning under MTP: Reverse Reasoning

We have seen that MTP significantly improves planning in language models. In this section, we investigate the underlying mechanisms of this improvement through a theoretical analysis of a two-layer disentangled Transformer on the star graph task. Going beyond disabling the Clever Hans cheat, we show that MTP induces a *reverse reasoning* process. All proofs in this section are deferred to Appendices C and D.

5.1 Problem Setup and Model

We focus on the 2-path, 3-node star graph with $N = T = 10$. Given the serialized graph and a query pair $(u_{\text{end}}, u_{\text{star}})$, the model must predict the path $(u_{\text{star}}, v, u_{\text{end}})$ autoregressively. The critical challenge is predicting the intermediate node v: this requires determining which path from u_{star} reaches u_{end}, a decision demanding genuine planning.

Input encoding. Let $\{e_i\}_{i=1}^N$ be the standard basis of $\mathbb{R}^N$. Given an ordered edge list $E = \{(u_1, v_1), \ldots, (u_M, v_M)\}$ and the query pair $(u_{\text{end}}, u_{\text{star}})$, content embeddings $z_{1:T} \in (\mathbb{R}^N)^T$ with $T = 2M + 2$ is

$$z_{2i-1} = e_{u_i},\ z_{2i} = e_{v_i}\ (i = 1, \ldots, M), \qquad z_{2M+1} = e_{u_{\text{end}}}, \quad z_{2M+2} = e_{u_{\text{star}}}.$$

The full input tokens are $x_t = (z_t; e_t) \in \mathbb{R}^{N+T}$. Each edge is encoded as two consecutive tokens; the last two tokens are u_{end} then u_{star}. Placing u_{end} before u_{star} ensures that u_{end} is visible in the prompt when the model predicts v, matching the autoregressive generation order. We write $t_{\text{end}}^{\text{ctx}}$ for the position of u_{end} in the context edge list, $t_v^{\text{ctx}} = t_{\text{end}}^{\text{ctx}} - 1$ for the position of v, and $t_{\text{end}}^{\text{prompt}} = T - 1$ for u_{end} in the prompt.

Training objective. We apply MTP with lookahead $k = 2$, targeting $y^{(1)} = v$ and $y^{(2)} = u_{\text{end}}$. Let $Z = [z_1, \ldots, z_T]^\top$ be the MTP context and $Z' = [z_1, \ldots, z_{T-1}, e_{y^{(1)}}]^\top$ the AR context (replacing z_T with $e_{y^{(1)}}$). The loss combines deep-head NTP terms with a shallow MTP head:

$$\mathcal{L}(G) = -\frac{1}{2}\left[\frac{1}{2}\Big(\underbrace{\log f_1(Z)e_{y^{(1)}}}_{\mathcal{L}_{1a}} + \underbrace{\log f_1(Z')e_{y^{(2)}}}_{\mathcal{L}_{1b}}\Big) + \underbrace{\log f_2(Z)e_{y^{(2)}}}_{\mathcal{L}_2}\right]. \tag{3}$$

The disentangled Transformer. To enable rigorous analysis, we adopt a simplified but expressive abstraction of causal attention and the Transformer. Following Zhang et al. (2024); Nichani et al. (2024); Huang et al. (2025), we merge the key and query matrices into a single matrix W.

Empirical motivation. We train two-layer disentangled transformers (one head per layer) comparing NTP and 2-MTP on this task. To isolate a minimal structure, we zero out cross-block entries in the query-key matrices and restrict each output head to read from a single

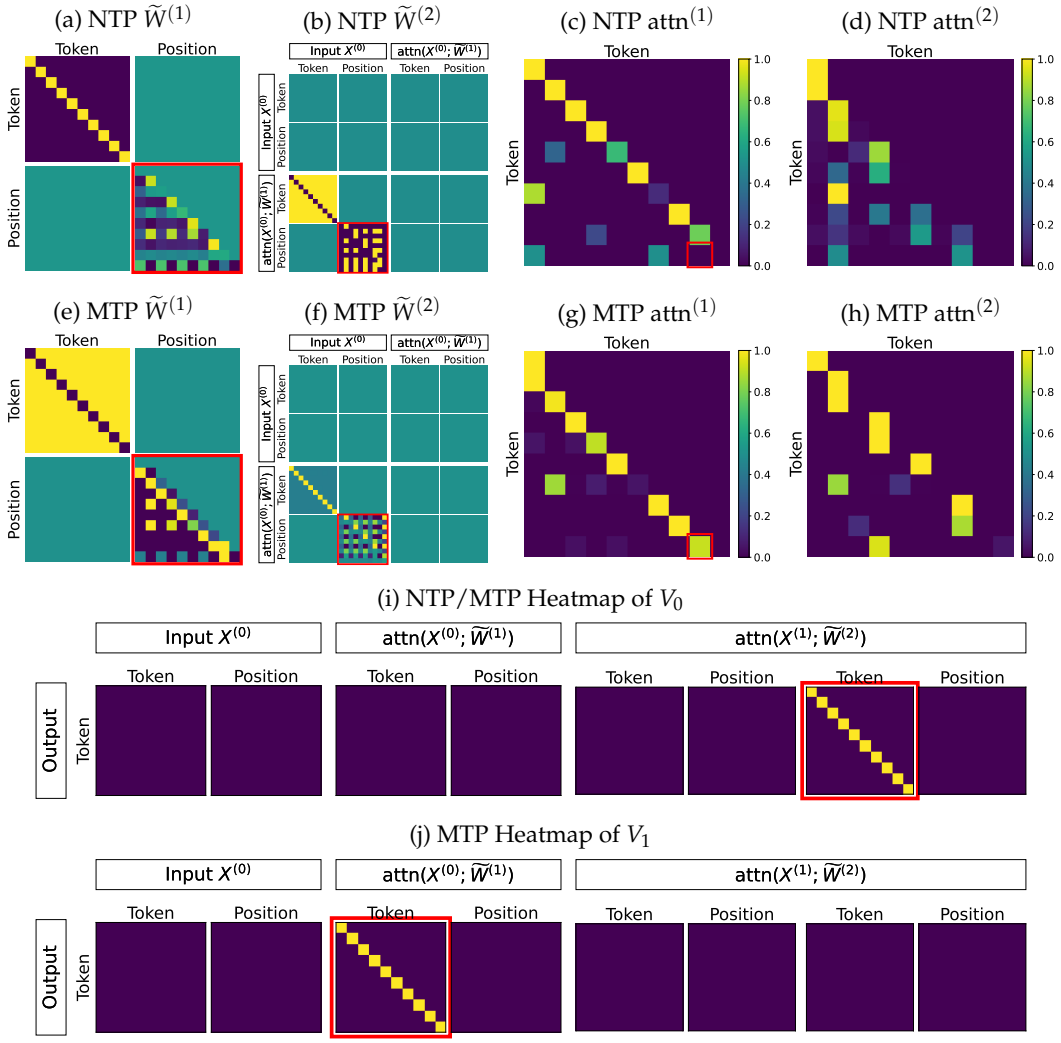


Figure 3: **Mechanism comparison between the minimal structures of NTP and 2-MTP on the star graph task.** (a–d) Query-key weights ($\widetilde{W}^{(1)}$, $\widetilde{W}^{(2)}$) and attention patterns (attn$^{(1)}$, attn$^{(2)}$) for NTP. (e–h) Corresponding weights and attention patterns for 2-MTP. Attention heatmaps are computed on a sequence with input graph $[3, 7, 6, 0, 7, 2, 3, 6, 0, 3]$ and target $[3, 6, 0]$. When predicting the difficult intermediate node, MTP (g) exhibits a clear predecessor-pointing pattern consistent with the reverse reasoning circuit (Theorem 1): Layer 1 attends sharply to the end node, whereas NTP (c) shows no such structure. (i) The output projection V_0 for NTP (identical in MTP) activates exclusively on the content block of the Layer 2 output. (j) The MTP-specific projection V_1 activates on the content block of the Layer 1 output. These block-sparse structures motivate the reduced model in Section 5.1.

content block—choices directly motivated by the learned weight structure (Figure 3). This minimal model preserves the core phenomenon: NTP fails while MTP succeeds. Under MTP, the model exhibits a clear *reverse reasoning* pattern: Layer 1 attends sharply to u_{end} when predicting v (Figure 3g), whereas NTP shows no such structure (Figure 3c). We refer to Appendix E.2 for the detailed setup and verify in Appendix E.3 that this reverse reasoning mechanism generalizes to *a larger, standard 8-layer 8-head Transformer on 5-path 5-node star graphs* with qualitatively identical attention patterns.

Definition 1 (Causal self-attention head). *For $W \in \mathbb{R}^{d \times d}$, define*

$$\text{attn}(X; W) := \text{softmax}(\text{MASK}(XWX^\top))\, X,$$

where $\mathrm{MASK}(M)_{i,j} = M_{i,j}$ *for* $i \geq j$ *and* $-\infty$ *otherwise.*

Definition 2 (Disentangled Transformer)**.** *Let* L *be the depth,* $d_1 = N + T$*, and* $d_\ell = 2d_{\ell-1}$ *for each* $\ell \in [L]$*. Let* $g_{1:T} = (g_1, \ldots, g_T)$ *denote the sequence of node identities. The input embeds each token as* $x_t = (z_t; e_t) \in \mathbb{R}^{d_0}$*, where* $z_t \in \mathbb{R}^N$ *and* $e_t \in \mathbb{R}^T$ *are one-hot content and position vectors, respectively. We then define* $X^{(0)} = [x_1, \ldots, x_T]^\top$*. For each* $\ell \in [L]$*, let* $\widetilde{W}^{(\ell)} \in \mathbb{R}^{d_\ell \times d_\ell}$ *denote the attention-weight matrix at layer* ℓ*. The model iterates*

$$X^{(\ell)} = \left[X^{(\ell-1)},\ \mathrm{attn}\left(X^{(\ell-1)}; \widetilde{W}^{(\ell)}\right)\right], \qquad \ell \in [L], \tag{4}$$

and its final output is $f_\theta(g_{1:T}) = X^{(L)}V$*.*

Guided by the empirical block structures of the attention weights and output heads in Figures 3a–b and 3i–j, respectively, we adopt the following simplified two-layer model. We set

$$\widetilde{W}^{(1)} = W^{(1)}, \qquad \widetilde{W}^{(2)} = \begin{pmatrix} 0 & 0 \\ W^{(2)} & 0 \end{pmatrix}, \tag{5}$$

and constrain

$$W^{(\ell)} = \begin{pmatrix} W_0^{(\ell)} & 0 \\ 0 & W_1^{(\ell)} \end{pmatrix}, \qquad W_0^{(\ell)} \in \mathbb{R}^{N\times N}, \quad W_1^{(\ell)} \in \mathbb{R}^{T\times T}, \qquad \ell \in \{1,2\}.$$

We further fix the two independent heads $V_0, V_1 \in \mathbb{R}^{d_2 \times N}$, which read from $X^{(2)}$, to the corresponding single-block selector forms. These simplifications preserve the NTP-fails/MTP-succeeds phenomenon, indicating that the underlying mechanism is captured by the attention dynamics. In particular, the block-diagonal parameterization yields a clean decomposition of the attention logits into content and positional terms (Lemma 1), enabling the gradient analysis in Sections 5.2–5.3.

Lemma 1 (Forward pass)**.** *Let* $Z = [z_1, \ldots, z_T]^\top$ *with* $z_t = e_{g_t}$ *denoting the content-only submatrix of the full input. Define*

$$A^{[\ell]} := Z\, W_0^{(\ell)} Z^\top + W_1^{(\ell)}, \qquad S^{[0]} := I_T, \qquad S^{[\ell]} := \mathrm{softmax}\big(\,\mathrm{MASK}(S^{[\ell-1]} A^{[\ell]})\big).$$

Then the output logits at the last position are $f_1(Z) = S^{[2]}_{[-1,:]} S^{[1]} Z$ *and* $f_2(Z) = S^{[1]}_{[-1,:]} Z$*.*

5.2 The Reverse Reasoning Circuit

We now characterize the attention configurations where the MTP loss achieves a stationary point. Rather than searching the full parameter space, we identify conditions on the softmax attention distributions $S^{[1]}$ and $S^{[2]}$ that simultaneously zero out all gradient terms.

Theorem 1. *The model achieves a stationary point for both the shallow loss* $\mathcal{L}_2$ *and the deep loss* $\mathcal{L}_1$ *if the attention distributions satisfy the following two conditions:*

1. ***Predecessor Pointing (Layer 1):*** *Layer 1 attends to the immediate predecessor at both the query position and the target context position:* $S^{[1]}_{T,:} = e_{T-1}^\top$ *and* $(S^{[1]})_{t^{\mathrm{ctx}}_{\mathrm{end}},:} = e_{t^{\mathrm{ctx}}_v}^\top$*.*
2. ***Content Matching (Layer 2, Last Row):*** *Layer 2 identifies the context position where* u_{end} *appears as part of an edge:* $S^{[2]}_{T,:} = e_{t^{\mathrm{ctx}}_{\mathrm{end}}}^\top$*.*

Note that a stationary point here means vanishing gradient, not necessarily zero loss. In particular, the AR component $\mathcal{L}_{1b}$ incurs nonzero error at this configuration, but its gradient is exponentially small (see Corollary 2 in Appendix D.1). This circuit is highly minimal. It relies strictly on *intra-edge backward attention*: forward attention within edges ($v \to u_{\mathrm{end}}$) is neither required nor utilized. Moreover, only two rows of $S^{[1]}$ are constrained (the query row and $t^{\mathrm{ctx}}_{\mathrm{end}}$); all other context rows are free, which is reflected in the sub-diagonal pattern being sharp at these two rows but diffuse elsewhere (Figure 3g). Having established the stationary manifold in terms of attention conditions, we now exhibit an explicit weight configuration that realizes it.

Corollary 1 (Constructive Existence of the Reverse Reasoning Circuit)**.** *Let $\gamma > 0$ be sufficiently large, and let $L \in \mathbb{R}^{T\times T}$ be the strictly lower shift matrix ($L_{i,i-1} = 1$, zero elsewhere). The following weight configuration satisfies both conditions of Theorem 1:*

- ***Layer 1 (Predecessor Shift):*** $W_0^{(1)} = 0, \quad W_1^{(1)} = \gamma\, L.$
- ***Layer 2 (Content Matching):*** $W_0^{(2)} = \gamma\, I$, $(W_1^{(2)})_{T-1,T-1} = -\gamma$ *(all other entries of $W_1^{(2)}$ are zero).*

Under this configuration, the loss and gradient norm satisfy $\mathcal{L} = O(e^{-\gamma})$ and $\|\nabla_\theta \mathcal{L}\| = O(e^{-\gamma})$.

Corollary 1 reveals a fundamental tension. The disentangled transformer possesses the representational capacity to solve the task perfectly—the reverse reasoning circuit lies within its hypothesis class. However, as we show in Section 5.3, pure NTP fails to discover this configuration from zero initialization. The failure is not one of expressivity but of *optimization*: NTP's gradient dynamics actively steer the model away from the predecessor pointer (Theorem 3), while MTP provides a clean two-phase gradient path that converges to this exact solution (Theorem 2). Comparing Figures 3c–d with 3g–h on the running example confirms this: MTP recovers the predecessor-pointing pattern in Layer 1 and sharp content matching in Layer 2, while NTP develops neither despite identical model capacity.

5.3 Why MTP Finds It and NTP Cannot

The reverse reasoning circuit exists within the model's hypothesis class (Corollary 1), yet whether gradient-based optimization discovers it depends critically on the training objective. The key mechanism is a *gradient decoupling* property: the shallow loss $\mathcal{L}_2$ routes gradients exclusively through Layer 1, entirely bypassing the uninitialized Layer 2. This enables a clean two-phase learning process under MTP, which we now formalize.

Theorem 2 (Cascaded Convergence to the Reverse Reasoning Circuit)**.** *Phase I: Consider the MTP loss $\mathcal{L}_2$ under gradient flow. Assume the content weight is fixed at $W_0^{(1)} = 0$, that $W_1^{(1)}$ has RoPE (Toeplitz) structure $W_{i,j} = w(i-j)$, where $w : \mathbb{Z} \to \mathbb{R}$ is a scalar function of the offset. Consequently, $S^{[1]}_{t,:} \to e_{t-1}^\top$ for all $t \geq 2$, i.e., Layer 1 converges to the universal predecessor pointer $S^{[1]} \to L$. Phase II: With $W_1^{(1)}$ frozen at γL. Then optimize $\mathcal{L}_{1a}$ over $(W_0^{(2)}, W_1^{(2)})$ from the zero initialization through gradient flow. Then, $W_0^{(2)} \to \gamma' I + B$, where $\gamma' \to +\infty$ and B is off-diagonal with finite negative entries $B_{d,u} < 0$ for $d \neq u$; $(W_1^{(2)})_{T-1,T-1} \to -\infty$; all other entries of $W_1^{(2)}$ remain at 0; and $S^{[2]}_{T,:} \to e_{t^{\text{ctx}}_{\text{end}}}^\top$.*

The two-phase structure is a direct consequence of MTP's gradient decoupling: Phase I uses $\mathcal{L}_2$, which bypasses Layer 2 entirely, to solve a *pure positional learning problem* for Layer 1. Phase II then benefits from a fully converged Layer 1, reducing the combinatorial task (jointly learn position + content matching) to a clean contrastive problem over content alone. Under pure NTP, this cascade cannot form—Layer 1 receives gradient only through the uninitialized Layer 2, leading to the misdirected gradient of Theorem 3.

Theorem 3 (Misdirected Gradient under Pure NTP)**.** *Consider the same setup as Phase I of Theorem 2: two-layer Disentangled Transformer, star graph task with $T = 10$, $W_0^{(1)} = 0$ fixed, $W_1^{(1)}$ with Toeplitz structure $w = 0$ at initialization, $W_0^{(2)} = W_1^{(2)} = 0$. Optimize the pure NTP deep loss. Then gradient descent* decreases *$w(1)$ (predecessor) and* increases *$w(k)$ for $k \geq 2$ (context): pure NTP actively repels the predecessor pointer and diffuses attention across context, structurally preventing formation of the reverse reasoning circuit.*

The intuition is straightforward. Under pure NTP with an uninitialized Layer 2, Layer 2 contributes a uniform $\frac{1}{T}$ weight to every position. The deep head's prediction is therefore an average over *all* rows of $S^{[1]}$, not just the query row. The target $y^{(1)} = v$ appears somewhere

in the context but *never* at the predecessor position $T-1$ (which encodes $u_{\text{end}} \neq v$). So the NTP gradient tells Layer 1: "attend to the context where the target might be"—exactly the opposite of the predecessor pointer that the 2-hop circuit requires. The shallow MTP loss avoids this entirely because it targets $y^{(2)} = u_{\text{end}}$, which *always* resides at $T-1$.

6 Conclusion

In summary, we show that the reasoning capabilities enabled by MTP arise from its impact on optimization dynamics. While prior work has established MTP's empirical advantages, our study strengthens these findings and, more importantly, formalizes the underlying mechanisms. By avoiding the entangled training signals of standard NTP, MTP leverages *gradient decoupling* to foster global planning abilities. We show that this training paradigm biases the network toward discovering interpretable algorithms, particularly a reverse reasoning process on graph path-finding tasks. Beyond validating these mechanisms across algorithmic and logic-based benchmarks, our work bridges a critical gap in understanding how training objectives shape model reasoning, providing a foundation for designing training paradigms for advanced reasoning.

Limitations and future work. Our theory focuses on a two-layer disentangled Transformer with a star graph task and lookahead $k = 2$. Extending to deeper architectures, general graph topologies, finite-time convergence rates, and sequential MTP variants (Liu et al., 2024) remains open.

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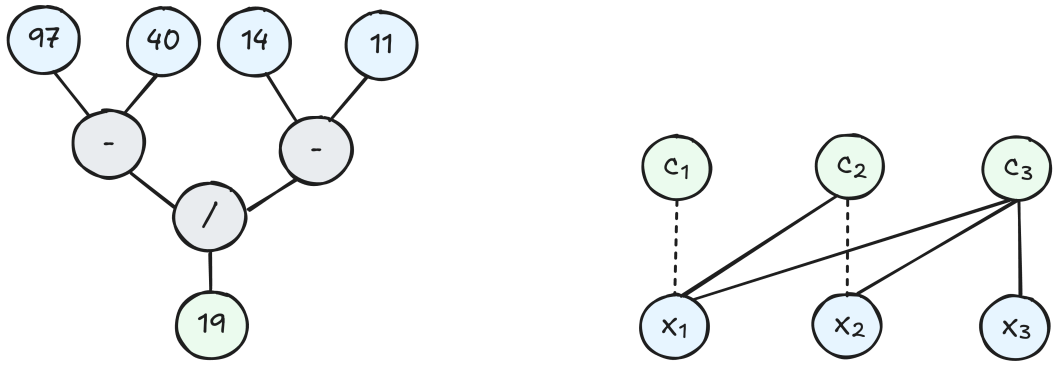
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A Additional Related Work of MTP

Foundations and Large-scale Integration. The standard NTP paradigm has driven the recent success of LLMs, yet it inherently suffers from limitations regarding planning. Early efforts to mitigate these pitfalls include ProphetNet (Qi et al., 2020), which proposed predicting future n-grams for sequence-to-sequence pre-training. Building on this, subsequent research identified the planning deficits of NTP and introduced teacherless, MTP style objectives (Bachmann & Nagarajan, 2024). This trajectory culminated in Gloeckle et al.



(a) Countdown (b) SAT

Figure 4: **Structural representations of reasoning tasks.** (a) Countdown: An expression tree showing the hierarchical computation to reach the target 19 using the numbers $\{11, 14, 40, 97\}$. The internal nodes represent arithmetic operators, and the root node represents the final result. (b) SAT: A bipartite factor graph representing a Boolean Satisfiability problem $(\neg x_1) \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee x_3)$. Nodes $\{x_1, x_2, x_3\}$ denote variables, and nodes $\{c_1, c_2, c_3\}$ denote logical clauses. Solid lines represent positive literals, while dashed lines indicate negated literals.

(2024), which formalized modern MTP frameworks and demonstrated that predicting multiple future tokens simultaneously can yield models that are both better at reasoning and significantly faster during inference.

Following its initial success, MTP has seen widespread adoption in state-of-the-art industry models to enhance both generation speed and performance. Technical reports for leading models such as DeepSeek-V3 (Liu et al., 2024), Qwen3-Next (Qwen Team, 2025), Nemotron 3 (Blakeman et al., 2025) and MiMo-V2-Flash (Xiao et al., 2026) highlight MTP as a core component of their training pipelines. Furthermore, the applicability of MTP has recently extended beyond text, showing significant promise in multi-modal architectures (Wang et al., 2025a).

Post-training Adaptation. While industrial models integrate MTP during the compute-intensive pre-training phase, an emerging line of research focuses on post-training adaptation. Specifically, several recent works (Liu et al., 2025; Gerontopoulos et al., 2025; Samragh et al., 2025) utilize fine-tuning strategies to efficiently transform standard NTP models into MTP models.

Future Prediction. Beyond standard MTP, a parallel line of research re-examines the fundamental modeling objective by shifting from point-wise prediction to holistic future modeling. At the sequence level, Mahajan et al. (2026) bypass discrete tokens entirely by pre-training models to predict high-level summaries. For finer-grained dependencies, Joint Token Prediction (JTP) (Ahn et al., 2025) and Parallel Token Prediction (PTP) (Will et al., 2026) move beyond independent token generation; while JTP enforces future-state compression via a representation bottleneck, PTP explicitly captures simultaneous token dependencies through latent variable modeling.

B Illustrations of Planning Tasks

In this section, we provide visual examples and formal definitions of the Countdown and SAT tasks used in our experiments. These tasks are designed to evaluate the model's ability to perform long-range planning and structural reasoning.

B.1 Countdown

Countdown is a mathematical planning puzzle where the model is given a set of source numbers and a target value. The goal is to construct a valid arithmetic expression using basic operators $(+, -, \times, /)$ to reach the target.

As illustrated in Figure 4a, the task can be represented as an expression tree. To solve this, a model cannot simply generate numbers greedily. It must internally plan the hierarchical structure of the computation as a single incorrect move would immediately render the target mathematically unreachable. For instance, to reach the target 19 from $\{11, 14, 40, 97\}$, the model must lock in the nested structure $(-)/(-)$ before outputting the first token. This ensures the very first operation is a committed step toward a globally valid solution.

B.2 SAT

The SAT task requires finding a truth assignment $\{0, 1\}$ that satisfies all clauses of a propositional formula in Conjunctive Normal Form (CNF).

As shown in Figure 4b, the task is represented as a bipartite factor graph where variables and clauses are interconnected. Within this structure, a single variable often appears in conflicting forms (e.g., x and $\neg x$) across multiple clauses, meaning any local assignment immediately propagates constraints throughout the entire network. Consequently, an initial choice that satisfies one local clause may preclude any valid assignment for the remaining variables, rendering the entire formula unsatisfiable. The model must therefore internally resolve these interconnected dependencies to ensure that the very first assignment remains compatible with a globally valid configuration.

C Technical Preliminaries

This appendix contains all lemma proofs, reorganised into three groups that mirror the logical structure of the argument.

Notation summary. For convenience, we collect the key symbols used throughout the appendix. After the block-diagonal factorisation in Lemma 1, the full input $X^{(0)}$ no longer appears; all subsequent analysis is expressed in terms of the content matrix Z.

Symbol	Definition	Space	Introduced
g_t	Node identity at position t	$[N]$	Definition 2
$x_t = (e_{g_t}; e_t)$	Full token (content + position)	$\mathbb{R}^{N+T}$	Definition 2
$z_t = e_{g_t}$	Content embedding at position t	$\mathbb{R}^N$	Lemma 1
$X^{(0)} = [x_1, \ldots, x_T]^\top$	Full input matrix (Lemma 1 proof only)	$\mathbb{R}^{T\times(N+T)}$	Definition 2
$Z = [z_1, \ldots, z_T]^\top$	Content matrix	$\mathbb{R}^{T\times N}$	Lemma 1
Z'	AR content context ($z_T \to e_{y^{(1)}}$)	$\mathbb{R}^{T\times N}$	Equation 3
$\widetilde{W}^{(\ell)}$	Full weight, layer ℓ	$\mathbb{R}^{2^{\ell-1}(N+T)\times 2^{\ell-1}(N+T)}$	Equation 4
$W_0^{(\ell)}$	Content weight, layer ℓ	$\mathbb{R}^{N\times N}$	Equation 5
$W_1^{(\ell)}$	Positional weight, layer ℓ	$\mathbb{R}^{T\times T}$	Equation 5
$A^{[\ell]} = ZW_0^{(\ell)}Z^\top + W_1^{(\ell)}$	Attention logit matrix, layer ℓ	$\mathbb{R}^{T\times T}$	Lemma 1
$S^{[\ell]}$	Attention distribution, layer ℓ; $S^{[0]} = I_T$	$\mathbb{R}^{T\times T}$	Lemma 1
$J(s) = \mathrm{diag}(s) - ss^\top$	Softmax Jacobian	$\mathbb{R}^{T\times T}$	
L	Strictly lower shift matrix ($L_{i,i-1} = 1$)	$\mathbb{R}^{T\times T}$	
$e_t \in \mathbb{R}^T$	Positional one-hot (selects row t)	$\mathbb{R}^T$	
$e_{g_t} \in \mathbb{R}^N$	Content one-hot (selects node g_t)	$\mathbb{R}^N$	

We follow the notation of Lemma 1 (main text): layers are 1-indexed, so Layer 1 has weight matrices $(W_0^{(1)}, W_1^{(1)})$ and produces attention distribution $S^{[1]}$, while Layer 2 has $(W_0^{(2)}, W_1^{(2)})$ and produces $S^{[2]}$. The initialisation $S^{[0]} := I_T$ is not a layer but the identity prior to any attention. Throughout, we write $J(s) := \mathrm{diag}(s) - ss^\top$ for the Jacobian of the softmax evaluated at a distribution s, $L \in \mathbb{R}^{T\times T}$ for the strictly lower shift matrix with $L_{i,i-1} = 1$, and I_T for the $T \times T$ identity.

- **Section C.1** establishes the forward-pass factorisation that underlies all subsequent gradient calculations (Lemma 1).
- **Section C.2** derives the gradient structure of the *deep* head $\mathcal{L}_1$, which couples both layers (Lemmas 2 and 3).
- **Section C.3** derives the gradient structure of the *shallow* MTP head $\mathcal{L}_2$ and proves the gradient decoupling property that is the key mechanism of MTP (Lemmas 4 and 5).

C.1 Forward-Pass Factorisation

Proof of Lemma 1. We specialize Definition 2 to $L = 2$ layers with one head per layer and unpack the disentangled structure.

By Definition 2, the input at position t is

$$x_t = \begin{bmatrix} z_t \\ e_t \end{bmatrix} \in \mathbb{R}^{d_0}, \qquad d_0 = N + T,$$

where $z_t = e_{g_t} \in \mathbb{R}^N$ is the one-hot content embedding and $e_t \in \mathbb{R}^T$ is the one-hot positional embedding. Stack these as $X^{(0)} = [x_1, \ldots, x_T]^\top \in \mathbb{R}^{T\times d_0}$.

Layer 1 attention logits. The disentangled assumption (Definition 2) means the attention weight matrix decomposes into content and positional blocks with no cross-terms:

$$W^{(\ell)} = \begin{bmatrix} W_0^{(\ell)} & 0 \\ 0 & W_1^{(\ell)} \end{bmatrix}, \qquad W_0^{(\ell)} \in \mathbb{R}^{N\times N},\ W_1^{(\ell)} \in \mathbb{R}^{T\times T}.$$

The (i,j)-entry of the raw attention logit matrix is therefore

$$\big(X^{(0)} W^{(1)} X^{(0)\top}\big)_{i,j} = x_i^\top W^{(1)} x_j = z_i^\top W_0^{(1)} z_j + e_i^\top W_1^{(1)} e_j.$$

The first term contributes $(Z W_0^{(\ell)} Z^\top)_{i,j}$ (content matching) and the second contributes $(W_1^{(\ell)})_{i,j}$ (positional bias, since $e_i^\top W_1^{(1)} e_j$ selects entry (i,j)). Hence

$$A^{[1]} := X^{(0)} W^{(1)} X^{(0)\top} = Z\, W_0^{(1)}\, Z^\top + W_1^{(1)} \ \in \mathbb{R}^{T\times T}.$$

Applying the causal mask and row-wise softmax:

$$S^{[1]} := \sigma\big(\mathrm{MASK}(A^{[1]})\big) \in \mathbb{R}^{T\times T}.$$

The Layer 1 attention output (Definition 1) is $\mathrm{attn}(X^{(0)}; W^{(1)}) = S^{[1]} X^{(0)} \in \mathbb{R}^{T\times d_0}$. Its content part at position t is

$$\big(S^{[1]} X^{(0)}\big)_{t,1:N} = S^{[1]}_{t,:}\, Z \ \in \mathbb{R}^{1\times N}. \tag{6}$$

The shallow (1-hop) prediction head reads the Layer 1 attention output at the last position T and projects onto the node vocabulary $[N]$. By equation 6:

$$\boxed{f_2(Z) = S^{[1]}_{T,:}\, Z \ \in \mathbb{R}^{1\times N}.}$$

Layer 2 attention logits. By Definition 2, $X^{(1)} = \left[X^{(0)},\ S^{[1]}\,X^{(0)}\right] \in \mathbb{R}^{T\times 2d_0}$. Under the disentangled architecture, Layer 2's attention query at position i is derived from the Layer 1 attention output $(S^{[1]}X^{(0)})_i$, while the keys remain the original input tokens x_j.[2]

The Layer 2 attention logit at (i,j) is then

$$\left(S^{[1]}X^{(0)}\right)_i^{\top} W^{(2)}\,x_j.$$

Applying the disentangled block-diagonal decomposition of $W^{(2)}$:

$$\left(S^{[1]}X^{(0)}\right)_i^{\top} W^{(2)}\,x_j = \left(S^{[1]}_{i,:}\,Z\right) W_0^{(2)}\,z_j \;+\; \left(S^{[1]}_{i,:}\right) W_1^{(2)}\,e_j.$$

In full matrix form, the content term gives $S^{[1]}\,Z\,W_0^{(2)}\,Z^{\top}$ and the positional term gives $S^{[1]}\,W_1^{(2)}$. Combining:

$$\text{Layer 2 logit matrix} = S^{[1]}\left(Z\,W_0^{(2)}\,Z^{\top} + W_1^{(2)}\right) = S^{[1]}\,A^{[2]}.$$

Applying the causal mask and row-wise softmax:

$$S^{[2]} := \sigma\big(\mathrm{MASK}(S^{[1]}\,A^{[2]})\big)\ \in \mathbb{R}^{T\times T},$$

which confirms the recursive formula $S^{[\ell]} = \sigma\big(\mathrm{MASK}(S^{[\ell-1]}\,A^{[\ell]})\big)$ with initial condition $S^{[0]} = I_T$.

The deep (2-hop) head reads from Layer 2's attention output at the last position. Layer 2 aggregates the Layer 1 attention outputs weighted by $S^{[2]}$, so the content part at position T is

$$S^{[2]}_{T,:}\left(S^{[1]}\,Z\right) = S^{[2]}_{T,:}\,S^{[1]}\,Z\ \in \mathbb{R}^{1\times N}.$$

Hence:

$$\boxed{f_1(Z) = S^{[2]}_{T,:}\,S^{[1]}\,Z\ \in \mathbb{R}^{1\times N}.}$$

Substituting definitions, we can verify the expanded expression. For the last row T, all positions $j \le T$ are visible under the causal mask, so the mask is vacuous for row T of $S^{[1]}A^{[2]}$:

$$\begin{aligned} S^{[2]}_{T,:} &= \sigma\big(S^{[1]}_{T,:}\,A^{[2]}\big) \\ &= \sigma\Big(\underbrace{\sigma\big(z_T^{\top}W_0^{(1)}Z^{\top} + W_1^{(1)}[T,:]\big)}_{S^{[1]}_{T,:}}\ \cdot\ \big(Z\,W_0^{(2)}\,Z^{\top} + W_1^{(2)}\big)\Big), \end{aligned}$$

and therefore

$$f_1(Z) = \sigma\Big(\sigma\big(z_T^{\top}W_0^{(1)}Z^{\top} + W_1^{(1)}[T,:]\big)\big(Z\,W_0^{(2)}\,Z^{\top} + W_1^{(2)}\big)\Big)\sigma\big(\mathrm{MASK}(Z\,W_0^{(1)}\,Z^{\top} + W_1^{(1)})\big)Z,$$

matching the stated expression. □

C.2 Gradient Structure of the Deep Head

The deep loss $\mathcal{L}_1 = -\log f_1(Z)\cdot e_{y^{(1)}}$ passes gradients through *both* Layer 1 and Layer 2. Lemma 2 identifies the three back-propagation paths; Lemma 3 makes these explicit for each parameter matrix.

[2]This is the property of the disentangled transformer with sparse parameters: each layer's query reads from the previous attention output, while the keys are always the original input tokens. This avoids the quadratic blowup from the concatenated representation and yields the clean recursive formula below.

Lemma 2. *Let $\mathcal{L}_1(Z) = -\log\Big(f_1(Z)\cdot e_{y^{(1)}}\Big)$ be the loss for the deep head evaluated on context $Z \in \{Z, Z'\}$ with target y. Let $S^{[1]}, A^{[2]}, S^{[2]}$ be the forward activations for Z. Then:*

$$\begin{aligned}
\mathrm{d}\mathcal{L}_1(Z) = &-\frac{1}{f_1(Z)\cdot e_{y^{(1)}}}\cdot\Big(S^{[1]}_{[-1,:]}\mathrm{d}A^{[2]}J(S^{[2]}_{[-1,:]})S^{[1]}\Big)Ze_{y^{(1)}}\\
&-\frac{1}{f_1(Z)\cdot e_{y^{(1)}}}\cdot\Big(\mathrm{d}S^{[1]}_{[-1,:]}A^{[2]}J(S^{[2]}_{[-1,:]})S^{[1]}\Big)Ze_{y^{(1)}}\\
&-\frac{1}{f_1(Z)\cdot e_{y^{(1)}}}\cdot\Big(S^{[2]}_{[-1,:]}\mathrm{d}S^{[1]}\Big)Ze_{y^{(1)}}.
\end{aligned}$$

Proof of Lemma 2. We apply the chain rule to $\mathcal{L}_1(Z) = -\log\big(f_1(Z)\cdot e_{y^{(1)}}\big)$ in following steps.

Outer log derivative.

$$\mathrm{d}\mathcal{L}_1 = -\frac{1}{f_1(Z)\cdot e_{y^{(1)}}}\,\mathrm{d}\big(f_1(Z)\cdot e_{y^{(1)}}\big).$$

Expand f_1. Substituting $f_1(Z) = S^{[2]}_{[-1,:]}\,S^{[1]}\,Z$ from Lemma 1:

$$\mathrm{d}\big(f_1(Z)\cdot e_{y^{(1)}}\big) = \mathrm{d}\Big(S^{[2]}_{[-1,:]}\,S^{[1]}\,Z\,e_{y^{(1)}}\Big).$$

Product rule over the two attention layers. $S^{[2]}_{[-1,:]}$ and $S^{[1]}$ are both functions of the parameters, so:

$$\mathrm{d}\Big(S^{[2]}_{[-1,:]}\,S^{[1]}\,Z\,e_{y^{(1)}}\Big) = \big(\mathrm{d}S^{[2]}_{[-1,:]}\big)\,S^{[1]}\,Z\,e_{y^{(1)}} \;+\; S^{[2]}_{[-1,:]}\,\big(\mathrm{d}S^{[1]}\big)\,Z\,e_{y^{(1)}}.$$

The second term is path (iii) directly. For the first term, we further expand $\mathrm{d}S^{[1]}_{[-1,:]}$ using the softmax Jacobian.

By the recursive formula $S^{[2]} = \sigma(\mathrm{MASK}(S^{[1]}\,A^{[2]}))$, the last-row logit vector is $\ell^{[2]} := S^{[1]}_{[-1,:]}\,A^{[2]}$ (the causal mask is vacuous for the last row). Its differential is

$$\mathrm{d}\ell^{[2]} = S^{[1]}_{[-1,:]}\,\mathrm{d}A^{[2]} + \mathrm{d}S^{[1]}_{[-1,:]}\,A^{[2]}.$$

Applying the softmax Jacobian $J(S^{[2]}_{[-1,:]})$:

$$\mathrm{d}S^{[2]}_{[-1,:]} = \mathrm{d}\ell^{[2]}\;J\big(S^{[2]}_{[-1,:]}\big) = \Big(S^{[1]}_{[-1,:]}\,\mathrm{d}A^{[2]} + \mathrm{d}S^{[1]}_{[-1,:]}\,A^{[2]}\Big)\;J\big(S^{[2]}_{[-1,:]}\big).$$

Substituting back and grouping by differential:

$$\begin{aligned}
\mathrm{d}\big(f_1\cdot e_{y^{(1)}}\big) = &\underbrace{S^{[1]}_{[-1,:]}\,\mathrm{d}A^{[2]}\;J\big(S^{[2]}_{[-1,:]}\big)\;S^{[1]}\,Z\,e_{y^{(1)}}}_{\text{path (i)}}\\
&+\underbrace{\mathrm{d}S^{[1]}_{[-1,:]}\;A^{[2]}\;J\big(S^{[2]}_{[-1,:]}\big)\;S^{[1]}\,Z\,e_{y^{(1)}}}_{\text{path (ii)}}\\
&+\underbrace{S^{[2]}_{[-1,:]}\,\mathrm{d}S^{[1]}\;Z\,e_{y^{(1)}}}_{\text{path (iii)}}.
\end{aligned}$$

Dividing by $-f_1(Z)\cdot e_{y^{(1)}}$ yields the stated expression. □

The three terms correspond to distinct backpropagation paths through the two-layer network.

Lemma 3. *Explicit gradients for $\mathcal{L}_1(Z)$ w.r.t. all four parameter matrices.*

$$\frac{\partial \mathcal{L}_1(Z)}{\partial W_0^{(1)}} = Z^\top \left[\frac{\partial \mathcal{L}_1}{\partial A^{[1]}}\right] Z. \qquad \frac{\partial \mathcal{L}_1(Z)}{\partial W_1^{(1)}} = \left[\frac{\partial \mathcal{L}_1}{\partial A^{[1]}}\right].$$

where $\frac{\partial \mathcal{L}_1}{\partial A^{[1]}} = M \odot \left[S^{[1]} \odot \left(\frac{\partial \mathcal{L}_1}{\partial S^{[1]}} - \frac{\partial \mathcal{L}_1}{\partial S^{[1]}} \odot S^{[1]} 1 1^\top\right)\right] \in \mathbb{R}^{T\times T}$ *is the loss gradient w.r.t. the Layer 1 logit matrix.*

$$\frac{\partial \mathcal{L}_1(Z)}{\partial W_0^{(2)}} = -\frac{1}{f_1(Z)\cdot e_{y^{(1)}}} \cdot Z^\top J(S^{[2]}_{[-1,:]}) S^{[1]} Z e_{y^{(1)}} S^{[1]}_{[-1,:]} Z.$$

$$\frac{\partial \mathcal{L}_1(Z)}{\partial W_1^{(2)}} = -\frac{1}{f_1(Z)\cdot e_{y^{(1)}}} \cdot J(S^{[2]}_{[-1,:]}) S^{[1]} Z e_{y^{(1)}} S^{[1]}_{[-1,:]}.$$

Proof of Lemma 3. We derive each gradient by applying the chain rule through $A^{[\ell]} = Z\, W_0^{(\ell)}\, Z^\top + W_1^{(\ell)}$.

Layer 1 parameters. Since $A^{[1]} = Z\, W_0^{(1)}\, Z^\top + W_1^{(1)}$, we have $\mathrm{d}A^{[1]} = Z\, \mathrm{d}W_0^{(1)}\, Z^\top + \mathrm{d}W_1^{(1)}$. By the chain rule and the trace identity $\mathrm{tr}(G\, Z\, \mathrm{d}W\, Z^\top) = \mathrm{tr}(Z^\top G\, Z\, \mathrm{d}W)$:

$$\frac{\partial \mathcal{L}_1}{\partial W_0^{(1)}} = Z^\top \frac{\partial \mathcal{L}_1}{\partial A^{[1]}} Z, \qquad \frac{\partial \mathcal{L}_1}{\partial W_1^{(1)}} = \frac{\partial \mathcal{L}_1}{\partial A^{[1]}}.$$

There are two paths into $S^{[1]}$

$$\frac{\partial \mathcal{L}_1}{\partial S^{[1]}} = -\frac{1}{f_1 \cdot e_{y^{(1)}}} \cdot \left[(S^{[2]}_{[-1,:]})^\top (Z e_{y^{(1)}})^\top + A^{[2]} J(S^{[2]}_{[-1,:]}) S^{[1]} Z e_{y^{(1)}} e_T^\top\right],$$

For each row r with mask $M_r = \mathrm{diag}(m_r)$, then

$$\frac{\partial \mathcal{L}_1}{\partial A^{[1]}_{[r,:]}} = \frac{\partial \mathcal{L}_1}{\partial S^{[1]}_{[r,:]}} J(s_1(r)) M_r.$$

Then the matrix format is

$$\frac{\partial \mathcal{L}_1}{\partial A^{[1]}} = M \odot \left[S^{[1]} \odot \left(\frac{\partial \mathcal{L}_1}{\partial S^{[1]}} - \frac{\partial \mathcal{L}_1}{\partial S^{[1]}} \odot S^{[1]} 1 1^\top\right)\right].$$

Layer 2 parameters. Only path (i) of Lemma 3 involves $A^{[2]}$ (paths (ii) and (iii) are independent of Layer 2 parameters). The relevant differential term is:

$$-\frac{1}{f_1 \cdot e_{y^{(1)}}}\, S^{[1]}_{[-1,:]}\, \mathrm{d}A^{[2]}\, \underbrace{J(S^{[2]}_{[-1,:]})\, S^{[1]}\, Z\, e_{y^{(1)}}}_{=: r \in \mathbb{R}^{T\times 1}}. \tag{7}$$

Gradient w.r.t. $W_0^{(2)}$. Substituting $\mathrm{d}A^{[2]} = Z\, \mathrm{d}W_0^{(2)}\, Z^\top$ into equation 7:

$$-\frac{1}{f_1 \cdot e_{y^{(1)}}} \underbrace{S^{[1]}_{[-1,:]} Z}_{1\times N}\, \mathrm{d}W_0^{(2)}\, \underbrace{Z^\top r}_{N\times 1}.$$

This is a scalar of the form $c^\top\, \mathrm{d}W\, d$ with $c = Z^\top (S^{[1]}_{[-1,:]})^\top$ and $d = Z^\top r$. Applying the trace identity $c^\top\, \mathrm{d}W\, d = \mathrm{tr}(d\, c^\top\, \mathrm{d}W)$:

$$\frac{\partial \mathcal{L}_1}{\partial W_0^{(2)}} = -\frac{1}{f_1 \cdot e_{y^{(1)}}}\, d\, c^\top = -\frac{1}{f_1 \cdot e_{y^{(1)}}}\, Z^\top r\, S^{[1]}_{[-1,:]} Z$$

$$= -\frac{1}{f_1 \cdot e_{y^{(1)}}}\, Z^\top J(S^{[2]}_{[-1,:]})\, S^{[1]}\, Z\, e_{y^{(1)}}\, S^{[1]}_{[-1,:]} Z.$$

Gradient w.r.t. $W_1^{(2)}$. Substituting $\mathrm{d}A^{[2]} = \mathrm{d}W_1^{(2)}$ into equation 7:

$$-\frac{1}{f_1 \cdot e_{y^{(1)}}} \underbrace{S^{[1]}_{[-1,:]}}_{1\times T} \mathrm{d}W_1^{(2)} \underbrace{r}_{T\times 1}.$$

Again of the form $c^\top \mathrm{d}W\, d$ with $c = (S^{[1]}_{[-1,:]})^\top$ and $d = r$. By the same trace identity:

$$\frac{\partial \mathcal{L}_1}{\partial W_1^{(2)}} = -\frac{1}{f_1 \cdot e_{y^{(1)}}}\, d\, c^\top = -\frac{1}{f_1 \cdot e_{y^{(1)}}}\, r\, S^{[1]}_{[-1,:]} = -\frac{1}{f_1 \cdot e_{y^{(1)}}}\, J\big(S^{[2]}_{[-1,:]}\big)\, S^{[1]}\, Z\, e_{y^{(1)}}\, S^{[1]}_{[-1,:]}. \quad \square$$

C.3 Gradient Decoupling of the Shallow MTP Head

The shallow MTP loss $\mathcal{L}_2$ is the key ingredient distinguishing MTP from NTP. Lemma 4 establishes the central gradient decoupling property: $\mathcal{L}_2$ routes gradients *exclusively* through Layer 1, entirely bypassing Layer 2. This provides a clean training signal for Layer 1 that does not depend on the (potentially uninitialised) Layer 2 weights—the mechanism enabling Phase I of Theorem 2. Lemma 5 makes this explicit per parameter matrix.

Lemma 4. *The gradient for the shallow head loss $\mathcal{L}_2 = -\log\Big(f_2(Z) \cdot e_{y^{(2)}}\Big)$ is:*

$$\mathrm{d}\mathcal{L}_2 = -\frac{1}{f_2 \cdot e_{y^{(2)}}} \cdot \Big(z_T^\top \mathrm{d}W_0^{(1)} Z^\top + \mathrm{d}W_1^{(1)}[-1,:]\Big) J(S^{[1]}_{[-1,:]}) Z e_{y^{(2)}}.$$

Proof of Lemma 4. We apply the chain rule to $\mathcal{L}_2 = -\log\Big(f_2 \cdot e_{y^{(2)}}\Big)$.

Outer derivative:

$$\mathrm{d}\mathcal{L}_2 = -\frac{1}{f_2 \cdot e_{y^{(2)}}}\, \mathrm{d}\big(f_2(Z) \cdot e_{y^{(2)}}\big).$$

Expand f_2: By Lemma 1, $f_2(Z) = S^{[1]}_{[-1,:]}\, Z$, so

$$\mathrm{d}\big(f_2(Z) \cdot e_{y^{(2)}}\big) = \mathrm{d}\big(S^{[1]}_{[-1,:]}\big)\, Z\, e_{y^{(2)}}.$$

Note that X is a fixed input matrix (not a function of parameters), so only $S^{[1]}_{[-1,:]}$ contributes a differential.

Softmax Jacobian. $S^{[1]}_{[-1,:]} = \sigma(A^{[1]}_{[-1,:]})$ where $A^{[1]}_{[-1,:]}$ is the last-row logit vector (the causal mask is vacuous for the last row). Therefore:

$$\mathrm{d}S^{[1]}_{[-1,:]} = \mathrm{d}A^{[1]}_{[-1,:]}\, J\big(S^{[1]}_{[-1,:]}\big).$$

Expand $\mathrm{d}A^{[1]}_{[-1,:]}$. From $A^{[1]} = Z\, W_0^{(1)}\, Z^\top + W_1^{(1)}$, the last row is $A^{[1]}_{[-1,:]} = z_T^\top\, W_0^{(1)}\, Z^\top + W_1^{(1)}[T,:]$. Differentiating with respect to the parameters:

$$\mathrm{d}A^{[1]}_{[-1,:]} = z_T^\top\, \mathrm{d}W_0^{(1)}\, Z^\top + \mathrm{d}W_1^{(1)}[T,:].$$

This involves only $(W_0^{(1)}, W_1^{(1)})$, confirming that $\mathcal{L}_2$ is independent of Layer 2.

Combining.

$$\mathrm{d}\mathcal{L}_2 = -\frac{1}{f_2(Z) \cdot e_{y^{(2)}}}\, \big(z_T^\top\, \mathrm{d}W_0^{(1)}\, Z^\top + \mathrm{d}W_1^{(1)}[T,:]\big)\, J\big(S^{[1]}_{[-1,:]}\big)\, Z\, e_{y^{(2)}}. \qquad \square$$

Note that $\mathrm{d}\mathcal{L}_2$ involves only the Layer 1 parameters $(W_0^{(1)}, W_1^{(1)})$ and is *entirely independent of Layer 2*. This is the central mechanism of MTP: the shallow head provides a direct, uncontaminated gradient signal to Layer 1 that does not need to backpropagate through the (potentially uninitialized or poorly conditioned) Layer 2 weights.

Lemma 5. *Explicit gradients for $\mathcal{L}_2$ w.r.t. all two parameter matrices.*

$$\frac{\partial \mathcal{L}_2}{\partial W_0^{(1)}} = -\frac{1}{f_2 \cdot e_{y^{(2)}}} \cdot z_T \, e_{y^{(2)}}^\top \, Z^\top \, J\big(S^{[1]}_{[-1,:]}\big) \, Z.$$
$$\frac{\partial \mathcal{L}_2}{\partial W_1^{(1)}} = -\frac{1}{f_2 \cdot e_{y^{(2)}}} \cdot \, e_T \, e_{y^{(2)}}^\top \, Z^\top \, J\big(S^{[1]}_{[-1,:]}\big).$$

Proof of Lemma 5. We route the differential from Lemma 4 through $A^{[1]} = Z\, W_0^{(1)}\, Z^\top + W_1^{(1)}$.

Step 1: Intermediate gradient $\partial \mathcal{L}_2 / \partial A^{[1]}$. Since only row T of $A^{[1]}$ affects the loss. Applying the softmax chain rule:

$$\frac{\partial \mathcal{L}_2}{\partial A^{[1]}_{T,:}} = -\frac{1}{f_2 \cdot e_{y^{(2)}}} \, e_{y^{(2)}}^\top \, Z^\top \, J\big(S^{[1]}_{[-1,:]}\big) \; \in \mathbb{R}^{1\times T},$$

and $\frac{\partial \mathcal{L}_2}{\partial A^{[1]}_{r,:}} = 0$ for $r \neq T$. In matrix form:

$$\frac{\partial \mathcal{L}_2}{\partial A^{[1]}} = -\frac{1}{f_2 \cdot e_{y^{(2)}}} \, e_T \, e_{y^{(2)}}^\top \, Z^\top \, J\big(S^{[1]}_{[-1,:]}\big) \; \in \mathbb{R}^{T\times T}. \tag{8}$$

Step 2: Gradient w.r.t. $W_1^{(1)}$. Since $A^{[1]} = Z\, W_0^{(1)}\, Z^\top + W_1^{(1)}$, the Jacobian $\partial A^{[1]}_{ij} / \partial (W_1^{(1)})_{ij} = 1$, so:

$$\frac{\partial \mathcal{L}_2}{\partial W_1^{(1)}} = \frac{\partial \mathcal{L}_2}{\partial A^{[1]}} = -\frac{1}{f_2 \cdot e_{y^{(2)}}} \, e_T \, e_{y^{(2)}}^\top \, Z^\top \, J\big(S^{[1]}_{[-1,:]}\big).$$

This is nonzero only in row T, since the prefactor e_T zeroes all other rows.

Step 3: Gradient w.r.t. $W_0^{(1)}$. Using the standard identity for bilinear forms ($A = Z\, W\, Z^\top$ implies $\partial f / \partial W = Z^\top \, (\partial f / \partial A) \, Z$):

$$\frac{\partial \mathcal{L}_2}{\partial W_0^{(1)}} = Z^\top \, \frac{\partial \mathcal{L}_2}{\partial A^{[1]}} \, Z = -\frac{1}{f_2 \cdot e_{y^{(2)}}} \, \underbrace{Z^\top e_T}_{=z_T} \, e_{y^{(2)}}^\top \, Z^\top \, J\big(S^{[1]}_{[-1,:]}\big) \, Z.$$

□

Together, Lemmas 3 and 5 provide the complete gradient landscape.

D Complete Proof of Theorems

D.1 The Reverse Reasoning Circuit

Proof of Theorem 1. By the chain rule, the gradient passing through any softmax distribution s is premultiplied by its Jacobian $J(s) = \mathrm{diag}(s) - s\, s^\top$. When s is a saturated one-hot vector e_k, the Jacobian evaluates to the zero matrix: $J(e_k) = 0$. We show that every active gradient chain passes through at least one such zero Jacobian.

By Lemma 5, $\nabla_{W^{(1)}} \mathcal{L}_2$ routes through $J(S^{[1]}_{T,:})$. Under Condition 1, $S^{[1]}_{T,:} = e_{T-1}^\top$ is one-hot, so $J(S^{[1]}_{T,:}) = 0$. Therefore $\nabla_{W^{(1)}} \mathcal{L}_2 = 0$.

By Lemma 3, both $\partial \mathcal{L}_1 / \partial W_0^{(2)}$ and $\partial \mathcal{L}_1 / \partial W_1^{(2)}$ contain the factor $J(S^{[2]}_{T,:})$. Under Condition 2, $S^{[2]}_{T,:} = e^\top_{t^{\text{ctx}}_{\text{end}}}$ is one-hot, so $J = 0$. Therefore $\nabla_{W^{(2)}} \mathcal{L}_1 = 0$.

By Lemma 3, the gradient $\nabla_{W^{(1)}} \mathcal{L}_1$ receives contributions from three paths:

- *Attention path (i) and query path (ii):* Both are gated by $J(S^{[2]}_{T,:})$, which vanishes under Condition 2 (same argument as case 2). Both paths are completely severed.
- *Value path (iii):* The error signal is $S^{[2]}_{T,:}\, \mathrm{d}S^{[1]}\, Z\, e_{y^{(1)}}$. Under Condition 2, $S^{[2]}_{T,:} = e^\top_{t^{\text{ctx}}_{\text{end}}}$, which selects only row $t^{\text{ctx}}_{\text{end}}$ of $\mathrm{d}S^{[1]}$. To reach the parameters $W^{(1)}$, this signal must pass through the row-wise softmax Jacobian $J\big((S^{[1]})_{t^{\text{ctx}}_{\text{end}},:}\big)$. Under Condition 1, $(S^{[1]})_{t^{\text{ctx}}_{\text{end}},:} = e^\top_{t^{\text{ctx}}_v}$ is one-hot, so this Jacobian is also zero. The value path is severed.

All three paths vanish, yielding $\nabla_{W^{(1)}} \mathcal{L}_1 = 0$.

Since Layer 2 selects only row $t^{\text{ctx}}_{\text{end}}$ via the one-hot $S^{[2]}_{T,:}$, all other context rows $r \neq t^{\text{ctx}}_{\text{end}}$ of $S^{[1]}$ receive zero error signal from $\mathcal{L}_1$. Combined with the vanishing of $\nabla_{W^{(1)}} \mathcal{L}_1$ (which only involves row T), these rows are gradient-free and therefore unconstrained at the stationary manifold. □

Proof of Corollary 1. We trace the forward pass under the proposed weight configuration for finite $\gamma > 0$ and show that every quantity deviates from its ideal (saturated) value by at most $O(e^{-\gamma})$.

Layer 1: Universal predecessor pointer. With $W_0^{(1)} = 0$, the content term vanishes and the logit matrix reduces to $A^{[1]} = W_1^{(1)} = \gamma\, L$. Under the causal mask, for each row $t > 1$ the entry at position $t-1$ has logit γ while all other visible entries have logit 0. The softmax therefore gives

$$S^{[1]}_{t,t-1} = \frac{e^\gamma}{e^\gamma + (t-1)} = 1 - \frac{t-1}{e^\gamma + (t-1)}, \qquad S^{[1]}_{t,j} = \frac{1}{e^\gamma + (t-1)} \quad (j \neq t-1). \tag{9}$$

Since $t \leq T$ is fixed, every off-diagonal entry is $O(e^{-\gamma})$ uniformly, so

$$S^{[1]}_{t,:} = e^\top_{t-1} + O(T\, e^{-\gamma}), \qquad \forall\, t > 1.$$

In particular, this approximately satisfies both parts of Condition 1:

- Row T attends to $T-1$: $S^{[1]}_{T,:} = e^\top_{T-1} + O(T\, e^{-\gamma})$.
- Row $t^{\text{ctx}}_{\text{end}}$ attends to $t^{\text{ctx}}_{\text{end}} - 1 = t^{\text{ctx}}_v$: $S^{[1]}_{t^{\text{ctx}}_{\text{end}},:} = e^\top_{t^{\text{ctx}}_v} + O(T\, e^{-\gamma})$.

Layer 2: Content matching with self-mask. By equation 9, the Layer 1 attention output at the query position T is

$$(S^{[1]}\, Z)_T = S^{[1]}_{T,:}\, Z = \big(1 - O(T\, e^{-\gamma})\big)\, e_{u_{\text{end}}} + O(T\, e^{-\gamma}).$$

Hence the effective content query for Layer 2 is $e_{u_{\text{end}}} + O(e^{-\gamma})$.

With $W_0^{(2)} = \gamma\, I$, the content logit from this query to key position j is

$$\big(e_{u_{\text{end}}} + O(e^{-\gamma})\big)^\top (\gamma\, I)\, z_j \;=\; \gamma\, \delta_{[z_j = e_{u_{\text{end}}}]} \;+\; O(\gamma\, e^{-\gamma}).$$

In the input sequence, u_{end} appears at exactly two positions: the context position $t^{\text{ctx}}_{\text{end}}$ and the prompt position $T-1$.

The positional mask $(W_1^{(2)})_{T-1,T-1} = -\gamma$ reduces the total logit at the prompt self-match $j = T-1$ from $\gamma + O(\gamma\, e^{-\gamma})$ to $O(\gamma\, e^{-\gamma})$, while the logit at $j = t^{\text{ctx}}_{\text{end}}$ remains $\gamma + O(\gamma\, e^{-\gamma})$. All other positions have logit $O(\gamma\, e^{-\gamma})$. Therefore the Layer 2 softmax at row T satisfies

$$S^{[2]}_{T,\,t^{\text{ctx}}_{\text{end}}} = 1 - O(T\, e^{-\gamma}), \qquad S^{[2]}_{T,j} = O(e^{-\gamma}) \quad (j \neq t^{\text{ctx}}_{\text{end}}),$$

which approximately satisfies Condition 2.

Deep-head output and loss bound. Composing both layers via Lemma 1:

$$\begin{aligned} f_1(Z)\cdot e_{y^{(1)}} &= S^{[2]}_{T,:}\, S^{[1]}\, Z\, e_{y^{(1)}} \\ &= \big(1 - O(T\, e^{-\gamma})\big)\, \big(S^{[1]}\, Z\big)_{t^{\text{ctx}}_{\text{end}}} \cdot e_{y^{(1)}} \;+\; O(T\, e^{-\gamma}) \\ &= \big(1 - O(T\, e^{-\gamma})\big)\, \big(e^{\top}_{t^{\text{ctx}}_{v}}\, Z\, e_{y^{(1)}} + O(T\, e^{-\gamma})\big) \;+\; O(T\, e^{-\gamma}). \end{aligned}$$

Since $e^{\top}_{t^{\text{ctx}}_{v}} Z\, e_{y^{(1)}} = z_{t^{\text{ctx}}_{v}} \cdot e_{y^{(1)}} = e_v \cdot e_v = 1$, we obtain

$$f_1(Z)\cdot e_{y^{(1)}} = 1 - O(T^2\, e^{-\gamma}).$$

Therefore, since $T = 10$ is fixed,

$$\mathcal{L}_{1a} = -\log\big(f_1(Z)\cdot e_{y^{(1)}}\big) = -\log\big(1 - O(e^{-\gamma})\big) = O(e^{-\gamma}).$$

An identical argument applied to $\mathcal{L}_2 = -\log\Big(f_2(Z)\cdot e_{y^{(2)}}\Big)$ yields $\mathcal{L}_2 = O(e^{-\gamma})$. Hence $\mathcal{L} = O(e^{-\gamma})$.

Every gradient chain in $\nabla_\theta \mathcal{L}$ passes through at least one row-wise softmax Jacobian $J(s) = \operatorname{diag}(s) - s\, s^{\top}$. Under the above weight configuration, each such s is $O(e^{-\gamma})$-close to a one-hot vector e_k, so

$$\|J(s)\| = \|J(e_k + O(e^{-\gamma}))\| = O(e^{-\gamma}),$$

since $J(e_k) = 0$ and the Jacobian is Lipschitz in its argument. All remaining factors in the gradient expressions (Lemmas 3 and 5) are bounded by polynomial functions of γ and T. Therefore

$$\|\nabla_\theta \mathcal{L}\| = O(e^{-\gamma}). \qquad \square$$

Corollary 2 (Forward Collapse of the AR Head)**.** *Under the same weight configuration as Corollary 1, the deep head evaluated on the AR context Z' outputs:*

$$f_1(Z') = e^{\top}_{y^{(1)}} \qquad \big(\textit{instead of its target } e^{\top}_{y^{(2)}}\big).$$

That is, the AR component of the MTP loss incurs nonzero error ($\mathcal{L}_{1b} > 0$), yet produces a vanishingly small gradient: $\|\nabla_\theta \mathcal{L}_{1b}\| = O(e^{-\gamma})$.

Proof of Collapse 2. Recall that Z' differs from Z only at the last position: $z'_T = e_{y^{(1)}}$ (i.e. e_v) instead of $e_{u_{\text{star}}}$.

Since Layer 1 uses pure positional bias ($W_0^{(1)} = 0$), the attention logits $A^{[1]} = W_1^{(1)} = \gamma\, L$ are content-independent. Replacing z_T does not change any entry of $A^{[1]}$, so the Layer 1 attention distribution on Z' coincides with that on X:

$$S'^{[1]}_{T,:} = S^{[1]}_{T,:} = e^{\top}_{T-1} + O(T\, e^{-\gamma}).$$

In particular, Layer 1 still retrieves the token at position $T-1$ as the effective query for Layer 2. Since $z'_{T-1} = z_{T-1} = e_{u_{\text{end}}}$ (position $T-1$ is unchanged), Layer 2 receives the same query $e_{u_{\text{end}}}$.

The prefix of Z' (positions $1, \ldots, T-1$) is identical to that of Z, so all context keys are unchanged. The Layer 2 logits and attention distribution therefore remain the same: $S'^{[2]}_{T,:} = S^{[2]}_{T,:}$. Composing both layers exactly as in the proof of Corollary 1:

$$f_1(Z') = S'^{[2]}_{T,:}\, S'^{[1]}\, Z' = e_{t_v^{\text{ctx}}}^\top Z' + O(e^{-\gamma}) = e_v^\top + O(e^{-\gamma}) = e_{y^{(1)}}^\top + O(e^{-\gamma}).$$

The AR target is $y^{(2)} = u_{\text{end}} \neq v = y^{(1)}$, so

$$f_1(Z') \cdot e_{y^{(2)}} = O(e^{-\gamma}) \quad \Longrightarrow \quad \mathcal{L}_{1b} = -\log\big(f_1(Z') \cdot e_{y^{(2)}}\big) = \gamma + O(1) > 0.$$

However, every gradient chain in $\nabla_\theta \mathcal{L}_{1b}$ passes through at least one softmax Jacobian $J(s)$ where s is $O(e^{-\gamma})$-close to a one-hot vector. As established in the proof of Corollary 1, each such Jacobian satisfies $\|J(s)\| = O(e^{-\gamma})$, so

$$\|\nabla_\theta \mathcal{L}_{1b}\| = O(e^{-\gamma}).$$

The error is effectively trapped: the loss is $\Theta(\gamma)$, but the gradient is exponentially small. □

D.2 Why MTP Finds It and NTP Cannot

Proof of Theorem 2. By Lemma 4,

$$\nabla_{W_1^{(1)}} \mathcal{L}_2 \;=\; -\frac{1}{f_2 \cdot e_{y^{(2)}}} e_T e_{y^{(2)}}^\top Z^\top J(S^{[1]}_{T,:}),$$

and this gradient is *entirely independent of* $W_0^{(2)}$ and $W_1^{(2)}$. Since only $\mathcal{L}_2$ is used in Phase 1, the Layer 1 weights exert no influence on the dynamics, regardless of their frozen values.

Under the Toeplitz constraint $W_{i,j} = w(i-j)$, all positions at the same offset k share the weight $w(k)$. Under random graph labeling, the expected gradient on $w(k)$ depends only on the offset k, so the Toeplitz structure is preserved exactly throughout Phase 1 (it is not merely approximately maintained).

$\mathcal{L}_2$ targets $y^{(2)} = u_{\text{end}}$, which is always at position $T-1 = 9$. With $T = 10$, the normalisation is $s_p + 8s_c + s_q = 1$. Let s_p be the softmax weight of the predecessor (offset $k = 1$), s_c the weight of each of the 8 context offsets $k \in \{2, \ldots, 9\}$, and s_q the self-weight. The target indicator satisfies $u_9 = 1$, $u_{10} = 0$, and $\mathbb{E}[u_t] = \frac{1}{8}$ for context positions $t \leq 8$.

The gradient flow equations for the Toeplitz weights are:

$$\dot{w}_p = -s_p + \frac{s_p}{s_p + s_c}, \tag{10}$$

$$\dot{w}_c = -s_c + \frac{1}{8} \cdot \frac{s_c}{s_p + s_c}. \tag{11}$$

The first term in each equation arises from the softmax normalisation Jacobian; the second term is the expected reward signal (the predecessor always hosts the target; each of the 8 context offsets hosts it with probability $\frac{1}{8}$).

Let $x = s_p/s_c \geq 1$. Subtracting equation 11 from equation 10:

$$\dot{\Delta} \;=\; -s_c(x-1) + \frac{x - \frac{1}{8}}{x+1}.$$

Using the normalization bound $s_c < \frac{1}{x+8}$ (since $s_q > 0$, with $T - 2 = 8$):

$$\dot{\Delta} > -\frac{x-1}{x+8} + \frac{x - \frac{1}{8}}{x+1}.$$

Cross-multiplying (both denominators positive) and expanding, positivity is equivalent to

$$\left(x-\tfrac{1}{8}\right)(x+8)-(x-1)(x+1) \;=\; x\underbrace{\left(8-\tfrac{1}{8}\right)}_{=\frac{63}{8}>0} \;=\; \frac{63x}{8} \;>\; 0$$

for all $x > 0$. Hence $\dot{\Delta} > 0$ globally and unconditionally for $T = 10$.

Note that at initialization $x = 1$ (i.e. $w = 0$), the gap derivative evaluates concretely to

$$\dot{\Delta}\big|_{x=1} \;=\; \frac{T-3}{2(T-2)}\bigg|_{T=10} \;=\; \frac{7}{16} \;>\; 0,$$

confirming that the predecessor pointer begins to emerge immediately from the zero initialization.

Since $\dot{\Delta} > 0$ uniformly, $w(1) - w(k) \to +\infty$ for all $k \geq 2$. Under the causal mask, row t of $A^{[1]} = W_1^{(1)}$ has a unique maximum at position $t-1$ (offset 1) as $\gamma := w(1) \to \infty$. Therefore:

$$S^{[1]}_{t,:} \;\to\; e_{t-1}^{\top} \qquad \forall\, t \geq 2.$$

With $T = 10$ this gives in particular: $S^{[1]}_{10,:} = e_9^{\top}$ (Condition 1) and $S^{[1]}_{t^{\text{ctx}}_{\text{end}},:} = e_{t^{\text{ctx}}_v}^{\top}$ for $t^{\text{ctx}}_{\text{end}} \in \{2,\ldots,8\}$ (Condition 1).

With $W_1^{(1)} = \gamma L$ ($\gamma \to \infty$) frozen, we have $S^{[1]}_{t,:} = e_{t-1}^{\top}$ for all $t \geq 2$. With $T = 10$, the effective Layer 1 query at the last position is

$$(S^{[1]}Z)_{10} \;=\; S^{[1]}_{10,:}Z \;=\; e_9^{\top}Z \;=\; z_9 \;=\; e_{u_{\text{end}}}.$$

The self-distractor (prompt position encoding u_{end}) is always at position $T-1 = 9$. The positional self-mask entry to be learned is therefore $(W_1^{(2)})_{9,9}$.

Since $S^{[1]} = L$ is frozen, the deep head output becomes $f_1 = S^{[2]}_{T,:}(LZ)$, where $(LZ)_j = z_{j-1}$. The target $y^{(1)} = v$ satisfies $z_{j-1} = e_v$ iff $j = t^{\text{ctx}}_v + 1 = t^{\text{ctx}}_{\text{end}}$. Hence:

$$L_{1a} \;=\; -\log S^{[2]}_{T,t^{\text{ctx}}_{\text{end}}},$$

a standard 1-hop contrastive problem over the Layer 2 attention distribution.

By gradient structure (from Lemma 3).

$$\nabla_{W_0^{(2)}} L_{1a} = -\frac{1}{f_1 \cdot e_{y^{(1)}}}\, Z^{\top} J(S^{[2]}_{T,:})\, e_{t^{\text{ctx}}_{\text{end}}}\, e_{u_{\text{end}}}^{\top}, \tag{12}$$

$$\nabla_{W_1^{(2)}} L_{1a} = -\frac{1}{f_1 \cdot e_{y^{(1)}}}\, J(S^{[2]}_{T,:})\, e_{t^{\text{ctx}}_{\text{end}}}\, e_{T-1}^{\top}. \tag{13}$$

Both are rank-1 outer products. Equation equation 12 modifies only row u_{end} of $W_0^{(2)}$; equation equation 13 modifies only row $T-1$ of $W_1^{(2)}$. Starting from $W_0^{(2)} = 0$ and $W_1^{(2)} = 0$, all other columns remain identically zero throughout Phase 2. This establishes Conclusion (i).

The logit from position 10 to position j in Layer 1 is:

$$\ell_j \;=\; \underbrace{e_{u_{\text{end}}}^{\top} W_0^{(2)} z_j}_{\text{content}} \;+\; \underbrace{(W_1^{(2)})_{9,j}}_{\text{positional}}.$$

The positional term indexes *row* 9 of $W_1^{(2)}$ (not row 10), because $S^{[1]}_{10,:} = e_9^{\top}$ shifts the effective query row by one. In particular, the self-mask entry is $(W_1^{(2)})_{9,9}$.

For a position k such that $z_k \neq e_{u_{\text{end}}}$ (a content distractor), the logit gap gradient is:

$$\dot{\Delta}_k \;=\; \dot{\ell}_{t^{\text{ctx}}_{\text{end}}} - \dot{\ell}_k \;\geq\; (1 - s_{j^*} - s_{\text{self}}) + s_k \;\geq\; 2s_k \;>\; 0,$$

where $j^* = t^{\text{ctx}}_{\text{end}}$ and the bound uses $1 - s_{j^*} - s_{\text{self}} \geq \sum_{k'} s_{k'} \geq s_k$ from the normalisation constraint. Thus all content distractor gaps diverge.

Position $k = 9$ ($= T - 1$) encodes $z_9 = e_{u_{\text{end}}}$ (same node as the query). Content updates to $W_0^{(2)}$ cancel for this position (query equals key). The gap is governed entirely by the positional bias $(W_1^{(2)})_{9,9}$:

$$\dot{\ell}_{j^*} = (1 - s_{j^*} - s_{\text{self}}) + (1 - s_{j^*}), \qquad \dot{\ell}_{\text{self}} = (1 - s_{j^*} - s_{\text{self}}) - s_{\text{self}}.$$

$$\dot{\Delta}_{\text{self}} \;=\; \dot{\ell}_{j^*} - \dot{\ell}_{\text{self}} \;=\; 1 - s_{j^*} + s_{\text{self}} \;>\; 0.$$

This drives $(W_1^{(2)})_{9,9} \to -\infty$, establishing Conclusion (iii).

From equation 12, the update to entry (i, u_{end}) of $W_0^{(2)}$ is

$$(\dot{W_0^{(2)}})_{i,u_{\text{end}}} = \frac{s_{t^{\text{ctx}}_{\text{end}}}}{\mu}\big(\delta_{i,u_{\text{end}}} - (Z^\top s)_{u_{\text{end}}}\big).$$

The diagonal entry ($i = u_{\text{end}}$) receives a positive update of $\frac{s_{j^*}(1-s_{j^*})}{\mu} > 0$ and diverges to $+\infty$. For off-diagonal entries $i \neq u_{\text{end}}$: as $W_0^{(2)} u_{\text{end}}, u_{\text{end}} \to +\infty$, the attention weight on non-u_{end} content positions vanishes, $(Z^\top s)_i \to 0$. Therefore $\int_0^\infty |(\dot{W_0^{(2)}})_{i,u_{\text{end}}}|\, dt < \infty$, and off-diagonal entries converge to a finite negative value $B_{i,u_{\text{end}}} < 0$. By graph-labeling symmetry, the same argument holds for every column, giving $W_0^{(2)} \to \gamma' I + B$ with $B_{ii} = 0$ and $B_{ij} < 0$ for $i \neq j$. This establishes Conclusion (ii).

□

Proof of Theorem 3. The full derivative of $\mu = \frac{1}{T}\sum_{r=1}^{T}(S^{[1]})_{r,:}\, u$ with respect to $w(k)$ is obtained by differentiating each row:

$$\frac{\partial \mu}{\partial w(k)} = \frac{1}{T}\sum_{r=k+1}^{T} \frac{\partial\big[(S^{[1]})_{r,:}\, u\big]}{\partial w(k)}.$$

(Rows $r \leq k$ have no position at offset k within their visible window and contribute zero.)

At $w \equiv 0$, row r has uniform attention $s_r = \frac{1}{r}1$ over positions $1, \ldots, r$. The Jacobian of softmax at s_r is $J(s_r) = \frac{1}{r}I - \frac{1}{r^2}11^\top$. Since $\frac{\partial A^{[1]}_{r,:}}{\partial w(k)} = e_{r-k}^\top$ (only entry $j = r - k$ is affected), the chain rule gives:

$$\frac{\partial\big[(S^{[1]})_{r,:}\, u\big]}{\partial w(k)} = \big(J(s_r)\, e_{r-k}\big)^\top u = \frac{1}{r}\left(u_{r-k} - \frac{1}{r}\sum_{j=1}^{r} u_j\right). \tag{14}$$

We take expectation over random graph labeling.

Rows $r \leq T - 2 = 8$. For these rows, $r - k \leq T - 3 \leq T - 2$, so $\mathbb{E}[u_{r-k}] = \frac{1}{T-2}$ for all $k \geq 1$. Also:

$$\mathbb{E}\left[\frac{1}{r}\sum_{j=1}^{r} u_j\right] = \frac{1}{r}\cdot\frac{r}{T-2} = \frac{1}{T-2}.$$

Therefore $\mathbb{E}\left[\frac{1}{r}\big(u_{r-k} - \frac{1}{r}\sum_{j=1}^{r} u_j\big)\right] = 0$.

Every row $r \leq T - 2$ contributes exactly zero in expectation. This is the symmetric regime: the target is equally likely at every context position, so no offset is favoured.

Row $r = T-1 = 9$. Here $\sum_{j=1}^{T-1} \mathbb{E}[u_j] = 1$ (since $\mathbb{E}[u_{T-1}] = 0$ and $\mathbb{E}[u_T] = 0$, all probability mass is on positions $1, \ldots, T-2$). For all offsets $k \geq 1$: $r - k = T - 1 - k \leq T - 2$, so $\mathbb{E}[u_{T-1-k}] = \frac{1}{T-2}$. Equation equation 14 gives:

$$\mathbb{E}\left[\frac{\partial\left[(S^{[1]})_{T-1,:}\, u\right]}{\partial w(k)}\right] = \frac{1}{T-1}\left(\frac{1}{T-2} - \frac{1}{T-1}\right) = \frac{1}{(T-1)^2(T-2)} > 0$$

for *all* $k \geq 1$. Row $T-1$ contributes the same positive term for every offset.

Row $r = T = 10$. Here $\sum_{j=1}^{T} \mathbb{E}[u_j] = 1$. Two sub-cases:

- $k = 1$ (predecessor offset): $r - k = T - 1$, so $\mathbb{E}[u_{T-1}] = 0$ (position $T-1$ encodes $u_{\text{end}} \neq v$). Equation equation 14 gives:
$$\frac{1}{T}\left(0 - \frac{1}{T}\right) = -\frac{1}{T^2} < 0.$$
- $k \geq 2$ (context offsets): $r - k = T - k \leq T - 2$, so $\mathbb{E}[u_{T-k}] = \frac{1}{T-2}$. Equation equation 14 gives:
$$\frac{1}{T}\left(\frac{1}{T-2} - \frac{1}{T}\right) = \frac{2}{T^2(T-2)} > 0.$$

Combining all three row regimes, the full expected gradient of μ is:

$$\mathbb{E}\left[\frac{\partial \mu}{\partial w(1)}\right] = \frac{1}{T}\left[\underbrace{0}_{\text{rows } r \leq T-2} + \underbrace{\frac{1}{(T-1)^2(T-2)}}_{\text{row } T-1} - \underbrace{\frac{1}{T^2}}_{\text{row } T}\right], \tag{15}$$

$$\mathbb{E}\left[\frac{\partial \mu}{\partial w(k)}\right] = \frac{1}{T}\left[\underbrace{0}_{\text{rows } r \leq T-2} + \underbrace{\frac{1}{(T-1)^2(T-2)}}_{\text{row } T-1} + \underbrace{\frac{2}{T^2(T-2)}}_{\text{row } T}\right] > 0, \quad k \geq 2. \tag{16}$$

From equation 15, the predecessor gradient has negative sign iff $(T-1)^2(T-2) > T^2$. Expanding:

$$(T-1)^2(T-2) - T^2 = T^3 - 5T^2 + 5T + 2.$$

For $T = 10$: $T^3 - 5T^2 + 5T - 2 > 0$. Therefore $(T-1)^2(T-2) > T^2$, giving:

$$\mathbb{E}\left[\frac{\partial \mu}{\partial w(1)}\right] = \frac{1}{T}\left(\frac{1}{(T-1)^2(T-2)} - \frac{1}{T^2}\right) < 0.$$

From $\mathcal{L}_1 = -\log \mu$ and the chain rule:

$$\mathbb{E}\left[\frac{\partial \mathcal{L}_1}{\partial w(k)}\right] = -\frac{1}{\mu_0}\mathbb{E}\left[\frac{\partial \mu}{\partial w(k)}\right].$$

Substituting equation 15 and equation 16:

$$\mathbb{E}\left[\frac{\partial \mathcal{L}_1}{\partial w(1)}\right] = \frac{1}{T\mu_0}\left(\frac{1}{T^2} - \frac{1}{(T-1)^2(T-2)}\right) > 0,$$

$$\mathbb{E}\left[\frac{\partial \mathcal{L}_1}{\partial w(k)}\right] = -\frac{1}{T\mu_0}\left(\frac{1}{(T-1)^2(T-2)} + \frac{2}{T^2(T-2)}\right) < 0, \quad k \geq 2.$$

At $T = 10$: $T - 1 = 9$, $T - 2 = 8$, $(T - 1)^2(T - 2) = 81 \times 8 = 648$.

$$\mathbb{E}\left[\frac{\partial L_0}{\partial w(1)}\right] = \frac{1}{10\mu_0}\left(\frac{1}{100} - \frac{1}{648}\right) = \frac{1}{10\mu_0} \cdot \frac{648 - 100}{64800} = \frac{548}{648{,}000\,\mu_0} > 0.$$

$$\mathbb{E}\left[\frac{\partial L_0}{\partial w(k)}\right] = -\frac{1}{10\mu_0}\left(\frac{1}{648} + \frac{2}{800}\right) = -\frac{1}{10\mu_0} \cdot \frac{800 + 2 \times 648}{648 \times 800} = -\frac{2096}{5{,}184{,}000\,\mu_0} < 0.$$

Gradient descent therefore:

- Decreases $w(1)$: the predecessor weight is *actively suppressed*.
- Increases $w(k)$ for $k \geq 2$: attention is *diffused* toward context positions.

This is precisely the opposite of the predecessor pointer $w(1) \to +\infty$, $w(k) \to -\infty$ for $k \geq 2$ required by the stationary manifold of Theorem 1.

□

E Additional Experiments Details

E.1 Experimental Setup for Section 4

In this section, we detail the experimental configurations for our standard multi-layer transformer models. All models are trained using the AdamW optimizer (Loshchilov & Hutter, 2019) with a weight decay of 0.

Graph. We conduct scaling experiments on 2-path 5-node star graphs and 5-node binary trees. To isolate the scaling effect of interest, when scaling one dimension (e.g., model capacity), we ensure the complementary dimension (e.g., dataset size) is sufficient to achieve performance saturation. Models are trained for 100 epochs with a batch size of 256 and a learning rate of 1×10^{-4}.

Remark 1. *For a fixed graph structure (e.g., the binary tree in Figure 1b), the set of possible problem instances is finite. With sufficient data (e.g., 0.5M samples) and large models (e.g., 10.65M parameters), NTP might memorize all possible problems and achieve perfect test performance. In this regime, the test set no longer reflects truly unseen problems. Thus, comparing MTP and NTP is more meaningful under limited data or model capacity, where memorization is unlikely and planning must be learned.*

Countdown. For the countdown task, we utilize a 4-level arithmetic setting featuring an 85M parameter model and a dataset of 1M samples. All numerical inputs and intermediate results are constrained to be less than 100. Training is performed for 100 epochs with a batch size of 1024 and a learning rate of 1×10^{-4}. Given that the model typically reaches peak performance early in the training process, we report the best performance achieved.

3-SAT. In the 3-SAT task, we focus on solving Boolean formulas featuring 7 variables and 45 clauses. We train a 5.3M parameter model on 50k samples for 300 epochs, using a batch size of 512 and a learning rate of 1×10^{-3}. We report the model's final performance at the conclusion of training.

E.2 Experimental Setup for Section 5

In this section, we detail the experimental setup for our 2-layer disentangled transformer models trained on 2-path 3-node star graphs. All models are trained using the AdamW optimizer with a batch size of 512 and a learning rate of 5×10^{-4} on 1M generated examples for 300 epochs.

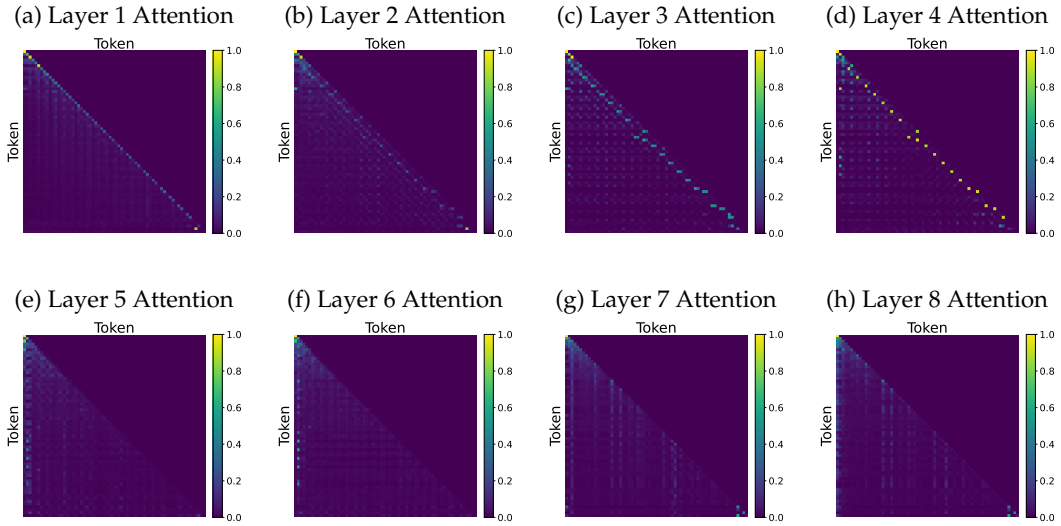


Figure 5: Attention heatmaps for the model trained with the standard NTP objective. The NTP model exhibits severe overfitting, achieving a training accuracy of 97% while the test accuracy only reaches 20%. Furthermore, the visualization demonstrates that the model primarily attends to the start node, specifically in Layers 7 and 8.

E.3 Empirical Validation on the Full Standard Architecture

In the main paper, our theoretical analysis relied on a disentangled mechanism and further mathematical simplifications to achieve tractability. In this section, we relax these structural assumptions and empirically evaluate whether our core findings generalize to a standard, full Transformer architecture.

Experimental Setup. We instantiate a standard 8-layer 8-head Transformer model with 25M parameters. The model is trained on a 5-path 5-node star graph dataset consisting of 1M training examples. We train the model for 40 epochs using a batch size of 1024 and a learning rate of 1×10^{-4}. An example of the data format is provided below:

Data Example

74,64 | 3,36 | 49,63 | 40,16 | 31,73 | 73,18 | 51,22 | 49,46 | 38,19 | 13,27 | 46,40 | 49,74 | 63,31 | 65,13 | 64,3 | 49,61 | 19,51 | 61,65 | 49,38 | 16,41 / 49,18 = 49,63,31,73,18

To understand the internal mechanisms driving the model's performance, we extract the attention maps specifically during the generation of the "difficult" token. The visualizations presented in this section are obtained by averaging the attention weights across 8 attention heads and 544 test examples.

Attention Analysis. We compare the standard NTP against 3-MTP. The results reveal a stark contrast in both generalization capability and internal attention routing.

As shown in Figure 5, the model trained with standard NTP suffers from severe overfitting. While it achieves 97% training accuracy, the test accuracy only reaches 20%. Analyzing the attention heatmaps, we observe that the NTP model primarily attends to the *start node*, specifically in Layers 7 and 8.

In contrast, Figure 6 illustrates the behavior of the model trained with 3-MTP. This model achieves perfect generalization, reaching 100% accuracy on both the training and test sets. Crucially, the attention mechanism operates entirely differently: rather than attending to the start node, the 3-MTP model identifies and heavily attends to the *end node*, predominantly in Layers 3 and 4.

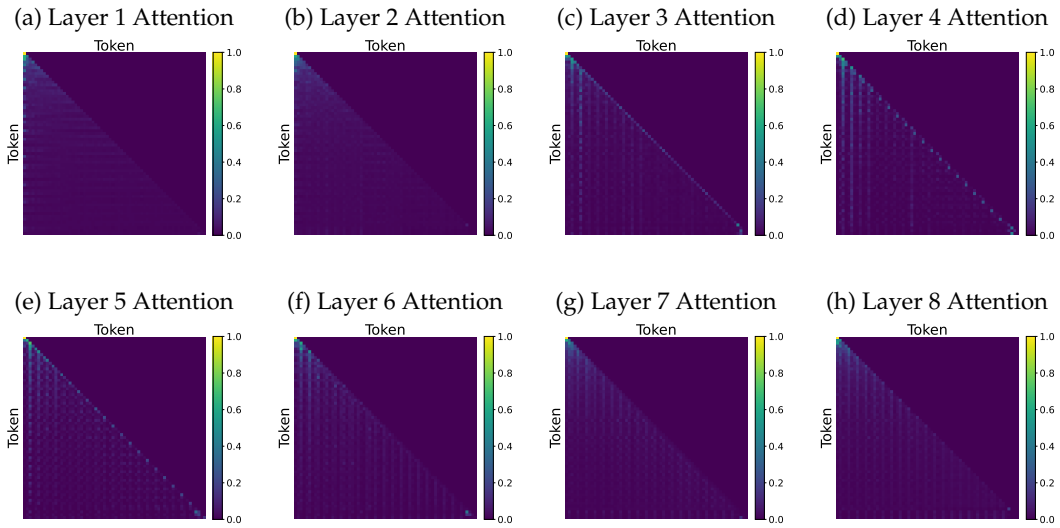


Figure 6: Attention heatmaps for the model trained with the 3-MTP objective. In contrast to the NTP model, the 3-MTP model achieves perfect generalization with 100% accuracy on both the training and test sets. Notably, the visualization highlights that the 3-MTP model primarily attends to the end node, predominantly in Layers 3 and 4.

These empirical results perfectly corroborate our theoretical findings. The observed structural shift in attention—specifically, prioritizing the end node in Layers 3 and 4—confirms that the MTP objective successfully induces a *reverse reasoning* mechanism, which is the key driver of the perfect generalization.